

Chain of Thought Reduces Errors from Misleading Names in Arithmetic

Anonymous ACL submission

Abstract

Asking language models to write out arithmetic calculations reduces answers that follow a misleading variable name. Building on Kudo et al.’s arithmetic study, we test thirteen models on pairs of problems with the same equations. In one version, a variable has an ordinary name; in the other, its name is a word for an incorrect digit. When asked only for the final answer, models more often return the digit suggested by the misleading name. Asking them to write the calculation before answering, known as chain-of-thought prompting, reduces this tendency. In 43 of 44 comparisons, the confidence interval for the added error rate lies within 2 percentage points of zero. Writing the calculation therefore makes answers more robust to misleading wording while the underlying math remains unchanged.

1 Introduction

Consider the problem

pen=1 + cup, cup=2 + 3; pen=?

pen has value 6. Renaming it four leaves the answer unchanged:

four=1 + cup, cup=2 + 3; four=?

Only the label changes: cup is still 5 and the queried variable is still 6. Answering 4 instead follows the name. We call 4 the *name-suggested value* and that answer a *name error*. Because 4 occurs nowhere in the equations and differs from every variable’s value, this error can be distinguished from copying a digit.

Can asking the model to calculate before answering reduce this error? In our *answer-only* condition, the model is prompted to return just the final digit. In our *chain-of-thought* (CoT) condition, it is prompted to restate and evaluate the equations before giving that digit. Here the calculation includes $\text{cup}=2+3=5$ and $\text{four}=1+5=6$. The comparison concerns what the model generates; an answer-

only response can still involve internal computation.

Misleading names already reduce accuracy on code tasks (Le et al., 2025; Lam et al., 2025; Le et al., 2026). We use arithmetic to make the competing answers explicit, building on Kudo et al. (2026)’s study of computation during CoT. We add a name that suggests a different answer while preserving the equations. This conflict, like those in Stroop and shape–texture studies (Stroop, 1935; Geirhos et al., 2019), separates the name’s suggestion from the computed value.

A separate analysis asks which values are readable inside the model. We train a *linear probe*, a small predictor of a variable’s true digit from internal activations, on separate problems. A probe predicting 6 and the model answering 6 are separate observations. We show that asking the model to write the calculation reduces answers matching the misleading name in almost every tested contrast. The paired renaming comparison makes resistance to a competing word a measurable benefit of CoT while holding the equations fixed.

2 Study design

We use Kudo et al.’s single-digit addition/subtraction task. The main level requires two operations and presents the queried variable’s equation first, before the equation needed to evaluate it. Two models cover all five levels (Appendix A). Matched versions change one name (Table 2): congruent names denote the true value; incongruent names denote another digit, absent from all inputs and variable values. We rename queried and intermediate variables separately. All names occupy one token, preserving matched token positions.

Both prompting conditions use three fixed sets of three worked examples. We evaluate thirteen Llama, Gemma and OLMo checkpoints (Appen-

Writing calculations reduces answers suggested by a misleading name Only the variable name changes:

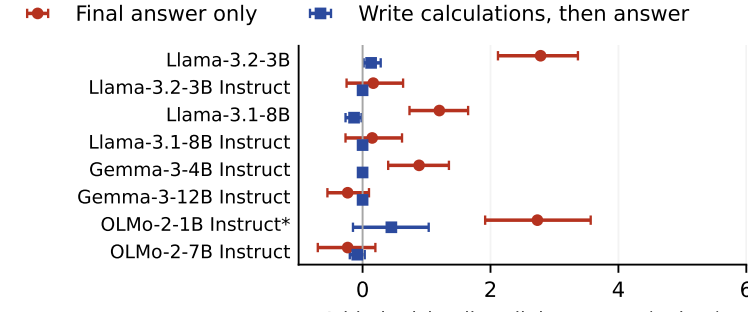
pen = 1 + cup rename four = 1 + cup
cup = 2 + 3 pen → four cup = 2 + 3
pen = ? four = ?

Correct answer stays 6.

Answering 4 follows the name.

4 is absent from the math.

(a) Same problems, two requested response formats



Added errors = renamed rate – ordinary-name rate. Plot: two-operation tasks; 95% intervals.

* Below the planned 90% neutral-calculation accuracy floor; included as a scope check.

(b) Across task depths and both variable roles

43 of 44

With written calculations, the added error rate is within 2 points of zero.

90% intervals. These are errors matching the wrong name, not all incorrect answers.

Figure 1: Renaming leaves the correct answer unchanged. (a) Writing calculations reduces added misleading-digit answers on two-operation tasks. (b) Across task depths and both variable roles, 43/44 written-calculation comparisons fall within the two-point margin. This concerns answers matching the misleading digit, not all errors.

080 dices A, F). The main table applies the preregis- 106
081 tered 90% mean neutral-CoT accuracy floor; ex- 107
082 cluded results appear in Table 4. 108

083 *Excess name errors* subtracts how often the 109
084 ordinary-name version answers the suggested digit 110
085 from how often the misleading-name version does. 111
086 This controls guesses when accuracy is low. *Fa-* 112
087 *cilitation* is congruent-minus-neutral accuracy; *in-* 113
088 *terference* is neutral-minus-incongruent accuracy. 114
089 Reliable effects have consistent signs across all 115
090 three demonstration sets and 95% intervals ex- 116
091 cluding zero. Equivalence requires a 90% inter- 117
092 val inside ± 2 points. Cluster bootstraps resam- 118
093 ple matched problems with their demonstration re- 119
094 peats (Appendix B). 120

095 Probes train on separate neutral problems with 121
096 three seeds. Random digits assigned to names 122
097 control for learning name identity (Hewitt and 123
098 Liang, 2019). Patching replaces name activations 124
099 with a matched twin’s (Zhang and Nanda, 2024); 125
100 neutral-word and alternative-name controls test 126
101 disruption and name specificity. Most CoT mea-
102 surements supply the correct calculation token by
103 token. Generated-chain probes select layers on
104 computation-disjoint validation problems before
105 held-out testing (Appendix N).

3 Results

3.1 Writing calculations limits misleading answers

Figure 1 shows added misleading-digit answers; Table 1 also reports changes in overall accuracy. Without CoT, a congruent name raises Llama-3.1-8B’s accuracy by +4.8 points, and a misleading name produces +1.2 points of excess name errors on the queried variable. Across levels, the largest effects occur where direct-answer accuracy is near chance; the matched baseline prevents those arbitrary digit answers from being attributed to the name.

With CoT, 43 of 44 name-error contrasts meet the equivalence bound. The exception is OLMo-2-1B-Instruct: its intermediate-variable excess is +3.2 points [+2.5, +3.9]. Its neutral CoT accuracy is 47%, compared with 77% under letter names. CoT limits answers matching the misleading digit. Overall accuracy still depends on names: 15 accuracy contrasts remain reliable (Appendix B).

3.2 What the probes recover

These probes read internal states while the correct calculation is supplied. Just after the intermediate equation, Llama-3.2-3B’s correct-digit

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
Gemma-3-12B-it	CoT	100 [100,100]	+0.0	+0.0	+0.0	+0.0
Gemma-3-12B-it	direct	78 [76,81]	-3.2*	-0.2	-3.7*	+0.4
Gemma-3-4B-it	CoT	100 [100,100]	-0.0	+0.0	-0.0	+0.0
Gemma-3-4B-it	direct	55 [48,59]	+0.2	+0.9*	+0.6	+0.3
Llama-3.1-8B-Instruct	CoT	100 [99,100]	+0.2	+0.0	+0.3	+0.0
Llama-3.1-8B-Instruct	direct	52 [45,65]	+1.7	+0.1	+5.2*	+0.2
Llama-3.1-8B	CoT	95 [86,100]	-1.3	-0.1	+3.8	-0.0
Llama-3.1-8B	direct	53 [41,62]	+4.8*	+1.2*	+3.8*	-0.0
Llama-3.2-3B-Instruct	CoT	100 [100,100]	-0.0	+0.0	-0.2	+0.0
Llama-3.2-3B-Instruct	direct	44 [40,50]	-1.8*	+0.2	+0.5	+0.5
Llama-3.2-3B	CoT	98 [97,100]	-1.2	+0.1	+0.9	-0.2*
Llama-3.2-3B	direct	22 [13,31]	+5.4*	+2.8*	+1.1	+1.8
OLMo-2-7B-Instruct	CoT	95 [93,97]	+0.7*	-0.1	-2.7	-0.0
OLMo-2-7B-Instruct	direct	54 [50,58]	+0.6	-0.2	+3.8*	-0.4

Table 1: Level 3 effects in percentage points: means across three demonstration sets [ranges]. *Helps*: congruent-name accuracy gain; *Errors*: excess name errors. *: consistent signs and 95% intervals excluding zero. CoT name-error contrasts meet the ± 2 -point equivalence bound.

accuracy is 49.0%, 31.8 points above the name-identity control. At the query, the control reaches 96.9%, showing that the probe can identify the name. Before the value is written, correct-digit accuracy reaches 97.9% and predictions of the name-suggested digit fall to zero. Gemma’s two variables show different readout timing (Appendix J).

Across 9 models, 17 of 18 variable readouts have neutral accuracy of 97.6–100.0% and excess name errors within 0.4 points of zero. OLMo-2-1B’s intermediate readout scores 21.8%. But the correlation between each model’s mean misleading-name probe accuracy and excess name errors reverses from $r = -0.997$ to $+0.776$ when OLMo-2-1B is omitted (Appendix L). Overall task competence remains a competing account. Within-problem fits also disagree on the association’s sign (Appendix E).

Probes trained on correct calculations score 19.4–62.2% on OLMo-2-1B’s neutral generated chains. Training on generated neutral chains raises this to 50.3–80.1%, still below the 90% minimum for interpreting these probes in every cell (Figure 2). Llama-3.2-3B reaches 96.3–100.0% with the same design, but its name-error cases remain rare. OLMo’s generated-chain readouts therefore

remain descriptive (Appendix N).

A generated calculation can state the correct value and still end with the suggested digit. Among matched versions that both state the correct value, OLMo-2-1B still has +3.6 points of excess name errors [+2.6, +4.7] across 2739 matched observations from 3 demonstration sets. The full control is in Appendix D.

3.3 Patching localizes the answer-only effect

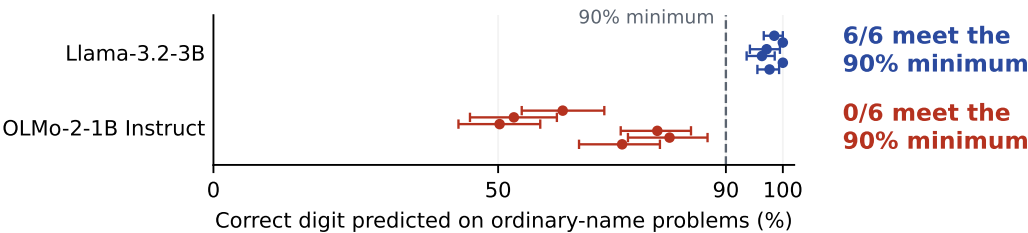
For Llama-3.2-3B without CoT, replacing name activations with the neutral twin’s removes 52% of name errors when all layers are patched (Figure 4). A span-by-layer sweep places the strongest effects at the defining equation in layers 0–7, where removal reaches 69% (Appendix K). The neutral-word control also damages 22% of correct answers. The patch identifies a region sensitive to the name; disruption limits claims about the restored computation.

The Llama-3.1-8B controls do not distinguish a name-specific cause: neutral and alternative-name patches remove errors similarly. Under a gold CoT, injecting the misleading name into a neutral run of the exception model moves 2.7% of answers to the name-suggested value, while the word con-

Predictors pass the accuracy check in Llama-3.2-3B, but fail in OLMo-2-1B

Separate predictor reads internal activity → predicts the variable’s correct digit.

Train and test on different ordinary-name problems, using calculations generated by the model.



Six points per model: 2 variables × 3 example sets; 95% intervals. Failed checks limit error-case interpretation.

Figure 2: A check on interpreting internal predictions. Predictors trained and tested on separate ordinary-name problems recover correct digits from generated calculations reliably in Llama-3.2-3B. All six OLMo-2-1B checks fall below the planned 90% minimum. These readouts therefore cannot yet explain its misleading-name errors.

trol changes 0.4% at the same layer. These interventions test only the selected models and positions; they do not establish a common causal mechanism across models.

4 Related work

Kudo et al. (2026) locate intermediate-value readouts during arithmetic CoT; our competing names distinguish a computed value from a lexical cue. Recent language-model Stroop studies test lexical responses (Wang, 2026; Hu et al., 2026). Faithfulness studies perturb prompts, chains or parameters (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025; Tutek et al., 2025); we combine a known conflicting cue with internal readouts. Positional copying and output priors can also explain arithmetic errors (Liu, 2026; Prabhakar et al., 2024); our copy control is reported in Appendix E. A companion code study (Anonymous, 2026) tests readouts before actual wrong value writes in Python traces and changes the demonstration format. Here the main finding is a reduction in arithmetic answers matching a misleading name while the equations remain unchanged.

5 Conclusion

Writing calculations limits answers suggested by misleading names in almost every tested arithmetic contrast. The paired renaming test measures resistance to a competing word while holding the equations fixed. The internal explanation remains unresolved: the probe association depends on one model, whose generated-chain readouts fail calibration even after retraining.

Limitations

- **Scope.** Short synthetic arithmetic with English number words does not establish real-code effects. Identifier-style names cover one level. Models are at most 12B, with raw completion prompts; chat templates and long-form reasoning are untested. Accuracy varies with demonstrations, so behavioral claims require agreement across three sets.
- **Internal measurements.** Probes establish decodability, not use. Most measurements supply correct chains. Generated-chain calibration passes for Llama-3.2-3B but fails for OLMo-2-1B, even after retraining (Appendix N); only two models are tested. The nine-model correlation is descriptive, reverses without OLMo, and confounds value availability with competence; variable roles are not independent models.
- **Interventions.** Patches replace mixed representations and damage correct answers. CoT also repeats names and equations; the value-only format changes performance and is not an equivalent intervention (Appendix H).

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Name	Problem (answer: 6)
Neutral	pen=1 + cup, cup=2 + 3; pen=?
Congruent	six=1 + cup, cup=2 + 3; six=?
Incongruent	four=1 + cup, cup=2 + 3; four=?

Table 2: Matched names preserve the arithmetic. The misleading digit, 4, occurs nowhere else.

Level	Problem (neutral names)	Steps	Held	Distr.
1	pen=1 + cup, cup=2; pen=?	1	1	0
2	pen=2 + 3, cup=1 + pen; cup=?	2	0	0
3	pen=1 + cup, cup=2 + 3; pen=?	2	1	0
4	pen=1 + cup, cup=2 + 3, box=4 + 5; pen=?	2	1	1
5	pen=1 + cup, cup=2 + box, box=1 + 2; pen=?	3	2	0

Table 3: The five levels of Kudo et al. (2026): additions or subtractions needed (*Steps*), variables unresolved when first read (*Held*), and unused equations (*Distr.*). At level 1 the intermediate value is stated rather than computed.

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A Task, models and replication

A problem contains single-digit variables, addition or subtraction, and a query. The five levels vary dependency order, depth and the presence of an unused equation (Table 3). Three same-level worked examples precede each test problem. The evaluated models are Gemma-3-12B-it, Gemma-3-4B, Gemma-3-4B-it, Llama-3.1-8B, Llama-3.1-8B-Instruct, Llama-3.2-3B, Llama-3.2-3B-Instruct, OLMo-2-1B-Instruct, OLMo-2-7B-Instruct, plus four derived checkpoints (Appendix F).

We reimplement Kudo et al. (2026)’s task and probe recipe rather than using their code. The shared choices are single-digit outputs, same-level demonstrations, a 10-way linear residual-stream classifier, full-batch SGD at 10^{-3} for 10,000 epochs, and their span-by-layer-window patching design. For replication we train on letter names and report t^* , t_{eq}^* , $\text{Acc}^{<\text{CoT}}$ and $\text{Acc}^{>\text{CoT}}$ at

$\tau = 0.9$. The Llama-3.2-3B value-writing positions agree with their Table 3: equation 5 for the queried variable and 2 for the intermediate. Our additions are the conflicting names, matched baselines, demonstration-set claim rule and equivalence bound.

On the letter condition, pre-chain probe maxima for Llama-3.2-3B are 30.4 and 47.9, compared with 17.8 and 33.2 reported by Kudo et al. (2026). The value-writing positions agree; the accuracy difference may reflect training details. Llama-3.1-8B’s intermediate value becomes decodable at equation 2, but its queried value becomes decodable at equation 3, earlier than Llama-3.2-3B’s equation 5. The larger model can therefore expose that value before writing its final evaluation. Full heatmaps appear in Appendix K.

B Equivalence checks

Following preregistration Amendment 3, equivalence requires a 90% cluster-bootstrap interval wholly inside the ± 2 -point margin (two one-sided 5% tests). We resample matched sets with their demonstration-seed replicates together, using 4,000 bootstrap draws. Other behavioral intervals remain at 95%. For CoT accuracy, 61 of 88 facilitation and interference contrasts meet the same equivalence bound. Of the 15 reliable accuracy contrasts, 5 are also equivalent within the margin and 10 are not; a detectable effect can still be small. The written-value control uses the same matched pairs for its excess estimate and interval; the exception model’s estimate pools all three demonstration sets and clusters on matched set.

C Tokenizer checks

We test candidate names in five tokenizer contexts and retain words that form one token without merging with a neighbor. For Llama 3, all ten number words and 95 of 112 neutral candidates pass. Gemma and OLMo use the same lists; runtime checks confirm equal token lengths and positions within matched groups. Spaces around binary operators prevent merges such as 1–two. No instance in the reported runs is excluded by the layout check.

D Additional naming and written-value controls

Written values. We split generated chains by whether they state the target’s true value. An ob-

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
OLMo-2-1B-Instruct	CoT	47 [44,51]	-5.7*	+0.4	+7.0*	+3.2*
OLMo-2-1B-Instruct	direct	34 [30,38]	-0.0	+2.7*	+3.0*	-0.2

Table 4: The level-3 model below the preregistered 90% mean neutral-CoT accuracy floor, excluded from the main behavioral table. Columns and claim markers follow Table 1.

Contrast	Cells	Equivalent	Reliable
CoT name errors	44	43	3
CoT accuracy	88	61	15
Direct name errors	44	31	13

Table 5: Equivalence within the two-point margin uses 90% intervals; reliability uses 95% intervals excluding zero and the same sign across all three demonstration sets. Each accuracy cell is one facilitation or interference contrast.

servation enters the excess estimate only when its neutral twin falls in the same split. The estimate and its 95% interval use those same paired observations. Pooled over eligible cells, the excess is +0.08 points in 132 cells where the value is stated and -0.10 in 27 where it is not. The exception persists in the stated-value split (main text), so this split does not explain its effect.

Identifier-style names. At level 3, four models receive names such as q4 instead of four, under three demonstration sets. Without CoT, queried-variable excess name errors range from -0.2 to +0.2 points; 0 of 4 cells are reliable. For number words on the same models, the range is -0.2 to +2.8, with 3 of 4 reliable. The numeric name-error effect is therefore stronger for number words in this comparison; congruent identifier names can still improve accuracy.

Word-class effects. With CoT at level 4, a number word on the queried variable costs Llama-3.2-3B 9.9 accuracy points whether the name agrees with the value or not. Those errors restate the wrong equation. This is distinct from answering the digit the name denotes, which is why the name-suggested value result cannot be read as removal of every naming effect.

Unused variables. A number word on the level 4 distractor changes accuracy by at most 1.2 points. Its no-CoT excess name errors range from +2.1 to +2.3; 2 of 2 cells are reliable. With CoT, the range is +0.0 to +1.0, with 0 of 2 reliable. At

the answer position, probe probability of the name-suggested value is 0.004 for a computed variable and 0.118 for the distractor; the latter’s true value has probe accuracy 0.04.

E Full sweeps and selectivity

Mixed-effects check. We compare the paired bootstrap with a mixed logit model, $\text{name_error} \sim \text{incongruent} + C(\text{seed})$, with a random intercept for matched set. Each twin’s outcome refers to the same name-suggested digit. Both procedures use the same fixed subsample of 600 sets per cell. Of 57 fitted cells, 50 agree on whether the interval excludes zero; 11 additional cells have no outcome variation. The mixed fit supports an effect in 5 cells where the bootstrap does not and withdraws support in 2 cells where it does (Gemma-3-12B-it, value-only chain, queried variable; Llama-3.1-Nemotron-Nano-8B, CoT, intermediate variable). Subsampling also removes support from 1 starred Table 1 cell under both procedures (Gemma-3-4B-it, no CoT, queried variable). The mixed model uses variational Bayes, so its interval is an approximate posterior interval. We retain the full-data bootstrap as the primary estimate on the reported percentage-point scale.

Instance-level association. Within each model, level, regime, variable and position, we regress name errors on the probe margin, $\log p_{\text{true}} - \log p_{\text{name}}$, where p_{name} is the probe probability of the name-suggested value, with standard errors clustered on matched set. The layer is selected by neutral accuracy. Of 63 fits, 29 have $p < .05$: 24 coefficients are negative and 5 positive. The direction is not consistent, and 4 positive results come from the model supplying most fits. This leaves the instance-level association unresolved. The model-level pattern is also sensitive to panel composition (Appendix L).

Name-error share. For comparison with shape-texture studies (Geirhos et al., 2019), we also

Target	Read	Errors	Neutral removed	Word dmg.	Alt. removed	Alt. followed
v1	value	0	–	1.6	–	0.0
v1	answer	0	–	1.6	–	0.0
v2	value	28	92.9	7.3	82.1	4.0
v2	answer	3	100.0	0.8	100.0	0.6

Table 6: OLMo-2-1B-Instruct, CoT, 500 matched pairs per target, patching name tokens at all layers. Removed: percentage of baseline name-suggested predictions removed by the neutral-twin patch. Word damage: percentage of correct neutral predictions damaged by the word control. Alt. removed: baseline name-suggested predictions removed by the alternative-name patch. Alt. followed: percentage following that source’s name-suggested value. Rates are percentages; – means undefined. These readouts force the gold chain, so their baseline errors differ from errors in the model’s own generated chains.

divide name errors by the total of name errors and correct answers, excluding congruent trials. Across 18 level 3 model–variable pairs, the index ranges from 0.02 to 0.38 without CoT and from 0.00 to 0.16 with CoT. The main text uses paired excess because the index does not subtract how often the neutral model answers that digit.

Positional-copy control. Across 4110 generated CoT name errors over models, levels and demonstration sets, the number immediately preceding the answer is the name-suggested value in 13% of cases and the target’s true value in 11%. A trailing name-suggested value is therefore absent from many errors. This check constrains the positional-copy explanation of Liu (2026), but does not identify what was copied in each case.

Probes report true-value accuracy, name-suggested prediction rate, probability of the name-suggested value and margin, with name-identity control accuracy and selectivity (Hewitt and Liang, 2019). P1 is the name token, P2 the end of its defining equation, P3 the query, P4 the position before its value is written, and P5 the position before the final answer. P1 is a reference, not evidence for a computed value. At P3 the control is near 1, so a readout of the name-suggested value there may reflect name identity. Figure 5 shows the CoT token sweep; Appendix K contains the other conditions and direct regime. Letter-trained replication uses the metrics in Appendix A.

F How the models were tuned

We compare six checkpoints on the Llama-3.1-8B architecture: pretrained, instruction-tuned, single-task LoRA, multi-task LoRA, TIES-merged adapters and reasoning-tuned. Adapters are folded

into the weights before evaluation. Shared tokenizers and identical matched problems permit paired comparisons (Table 7). A pretrained/instruct Gemma pair supplies a second family comparison.

Tuning	Model	Queried	Intermediate	Acc.
base	Llama-3.1-8B	+1.20*	-0.02	53.4
instruct	+ instruct	+0.15	+0.22	52.2
single-task finetune	+ task LoRA	+0.05	+0.23	52.2
multi-task finetune	+ multi-task LoRA	-0.58*	-0.08	46.1
weight merge	+ TIES merge	-0.10	+0.10	51.9
reasoning-tuned	Nemotron-Nano-8B	+1.35*	+0.33	39.4
base	Gemma-3-4B	+0.08	+0.02	60.2
instruct	+ instruct	+0.88*	+0.33	54.9

Table 7: Level 3, no CoT, name errors above the matched twin’s rate, mean over three demonstration sets. Acc. is neutral accuracy. * as in Table 1. The bottom block repeats the pretrained-to-instruct step in a second family, Gemma-3-4B.

The pretrained and reasoning-tuned Llama checkpoints have reliable positive queried-variable excess name errors without CoT (+1.20 and +1.35 points). The three task-tuned variants differ from the instruction-tuned checkpoint by at most 0.73 points across 6 comparisons. These are observed differences, not tests of checkpoint-to-checkpoint equivalence. Every checkpoint’s CoT excess is at most 0.20 points in magnitude.

Task accuracy complicates interpretation. The multi-task model’s reliable negative excess (-0.58) accompanies lower neutral accuracy (46.1%, versus 52.2%). The reasoning-tuned checkpoint has the lowest accuracy (39.4%). It also requires a system instruction to enable long-form reasoning; our raw-completion prompts leave that mode off. The result concerns the checkpoint under this prompt, not reasoning mode in deployment.

Only the instruct Gemma effect passes the claim rule (+0.88, versus +0.08 pretrained). The difference is untested and both estimates lie within the

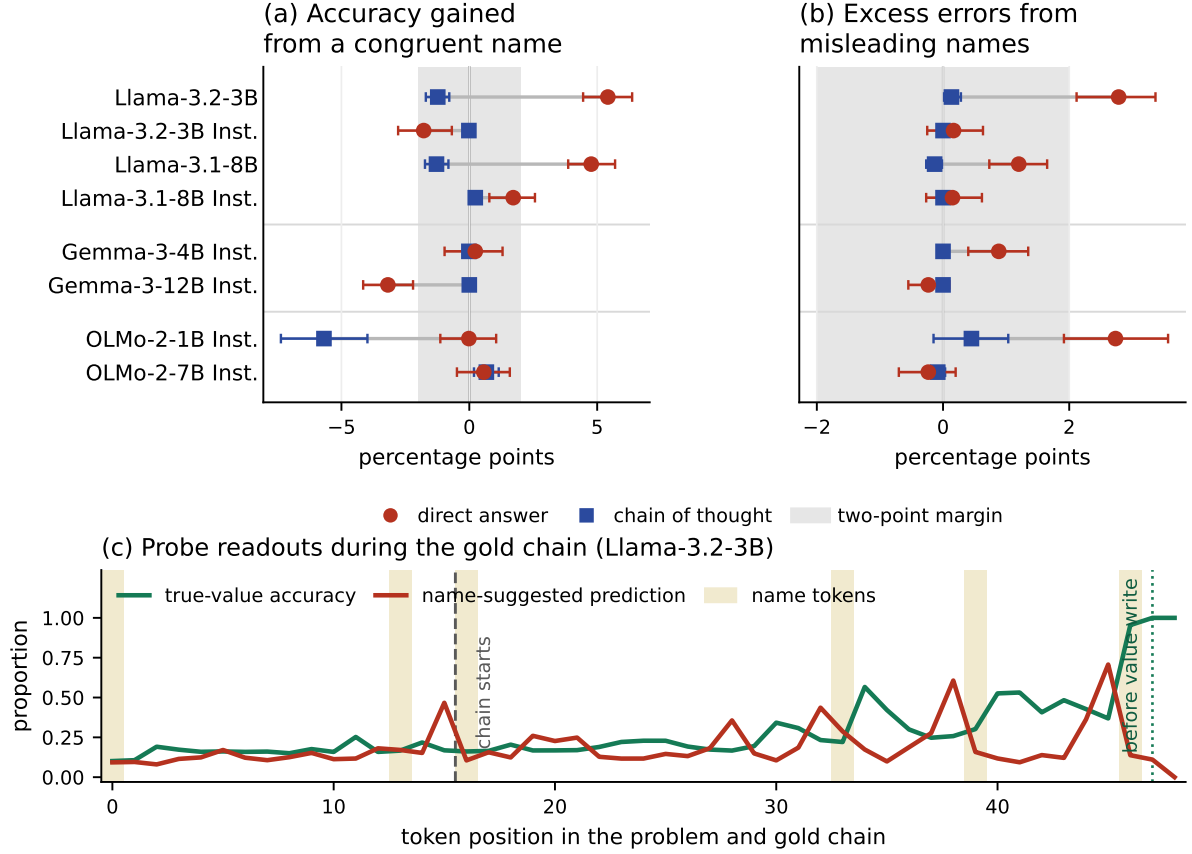


Figure 3: Level 3 queried-variable effects: **(a)** congruent-name accuracy gains; **(b)** excess name errors. Means and 95% intervals across three demonstration sets; grey: ± 2 points, a reference band rather than an equivalence test. **(c)** Best-layer true-value accuracy and name-suggested predictions under a supplied correct calculation. OLMo-2-1B is below the planned neutral-accuracy floor. Shading: name positions; dotted line: position before the queried value.

margin, so the pair cannot establish where in training name interference arises.

G Probe recipe and patching scope

We check two implementation choices inherited from Kudo et al. (2026): the probe optimizer and the name occurrences included in a patch.

Recipe. We refit the level-3 probes for Llama-3.2-3B with L-BFGS, and with SGD on standardized inputs, against the reported recipe.

Position	Variable	SGD (reported)	L-BFGS	standardized
defining equation	queried	0.184	0.190	0.183
defining equation	intermediate	0.496	0.422	0.420
query	queried	0.183	0.316	0.280
query	intermediate	0.271	0.366	0.393
value step	queried	1.000	1.000	1.000
value step	intermediate	0.982	0.962	0.972
answer	queried	1.000	1.000	1.000

Table 8: Best-layer true-value accuracy on neutral problems under three probe recipes (proportions from 0 to 1).

All three recipes give the same qualitative result: the value is decodable where the chain writes it (0.96–1.00) and not at the end of its defining equation (0.18–0.50). The query position moves, from 0.271 under the reported recipe to 0.393 under standardization, within a range of 0.18–0.39 over recipes and variables. We already report that position as the presence of the name rather than a decoded value, because its control task scores 0.97, so this variation does not change the interpretation of that position.

Scope. Patching every occurrence of the name includes the copy the model writes in its own output. In the direct regime, restricting the patch to the name’s occurrences in the prompt leaves the intermediate variable essentially unchanged, since its name never appears in the answer: removal is 43% at layers 0–3 under both scopes. For the queried variable the best windows remove similar shares (52% at layers 4–7 prompt-only, against

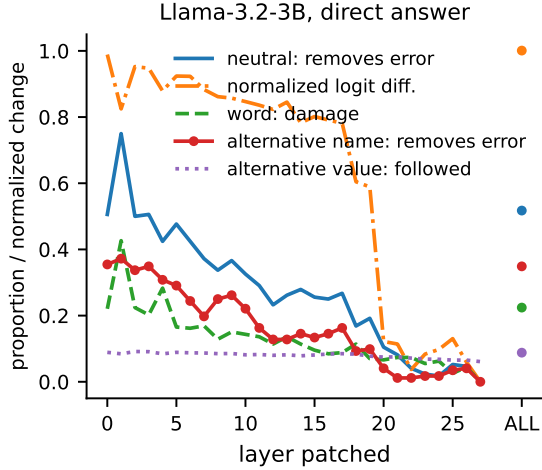


Figure 4: Llama-3.2-3B, answer-only: queried-variable name errors removed by layer, with neutral-word damage and alternative-name controls. Removal counts changes away from the name-suggested digit.

53% at layers 0–3), the prompt-only patch removes less in the later windows, and the two disagree most in the first window, where the prompt-only patch removes 3% and damages 76% of correct answers. That is mechanical: the queried variable’s name also sits at the position we read, so patching its prompt copies alone leaves the model reading a name whose early representation no longer matches. The localization we report does not depend on the choice.

H A chain that writes values without equations

Our chain restates each equation before substituting into it, as Kudo et al.’s does. Their Tables 5–6 also report a format that writes only the sub-results, `cup=5`, `pen=6`. That format is the arithmetic counterpart of a code trace, which writes a variable’s value and never the expression that produced it, so we ran it on the panel at level 3 with three demonstration sets, with two predictions registered in advance for both Llama models: that the excess name errors at the intermediate’s value step would exceed the 2-point margin, and that the value would still be decodable there (probe accuracy at least 0.85), making any failure one of readout.

Both predictions fail as registered. Llama-3.1-8B writes the name-suggested value at the intermediate’s value step +0.15 points above its twin, inside the margin. Llama-3.2-3B is at +9.20, which

passes our claim rule, but across the three demonstration sets that is +1.7 / +24.2 / +1.7, so one set carries most of it. More informative is what the probes say. At the step that writes the value, the true value is decodable at 0.81 and 0.59 for Llama-3.1-8B and at 0.61 and 0.79 for Llama-3.2-3B, against 1.00 and 0.99 for the same model under the full chain. Neutral accuracy falls with it, to 54% and 31%.

Removing the equations lowers both accuracy and true-value probe readouts. It therefore fails to isolate a readout error while preserving those measures. This result does not support a shared explanation for value-only arithmetic chains and code traces.

I Additional levels, models and demonstration sets

Table 9 repeats Table 1 for the other four levels. The pretrained Gemma-3-4B, the base half of the Gemma pair, is reported in Table 7 rather than in Table 1.

Data, compute and software. Each level has 2,000 matched sets, each rendered in every condition, and 10,000 separately generated problems for training probes; every SGD probe is trained with 3 seeds. Every run here is inference and the probes are linear; the three adapter checkpoints of Appendix F were trained in a separate project, and that compute is not counted below. The recorded allocation time totals at most 185 GPU-hours on NVIDIA A30, H100 and H200 cards, including idle time and shared workers that also ran companion-code experiments. The models are from the Llama 3 (Grattafiori et al., 2024), Gemma 3 (Gemma Team, 2025) and OLMo 2 (Team OLMo, 2024) families, with the reasoning-tuned checkpoint from Bercovich et al. (2025), each used under its release licence for research. Models run in PyTorch (Paszke et al., 2019) with Transformers (Wolf et al., 2020), and the probes are trained in PyTorch; the L-BFGS refit of Appendix G uses scikit-learn (Pedregosa et al., 2011), and the mixed-effects check uses statsmodels (Seabold and Perktold, 2010).

Per-set values, the pooled estimate and the claim rule are released with the code, together with the level-4 irrelevant-name-suggested value control, the name-suggested value-distance analysis, the teacher-forced name-suggested prediction

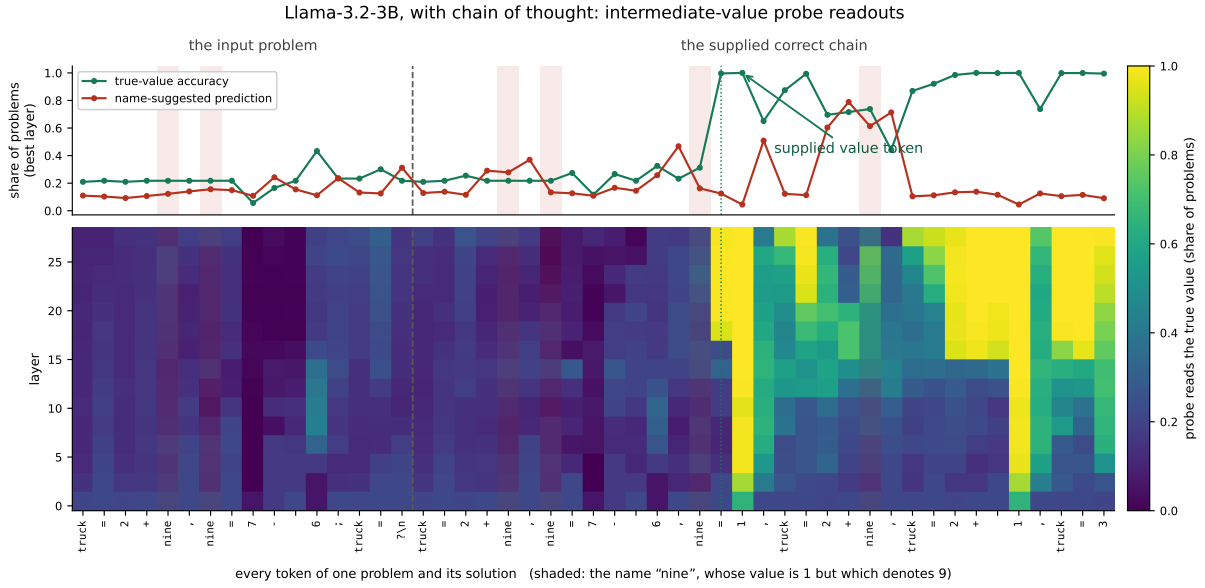


Figure 5: In the layout of [Kudo et al.](#)’s Figure 2. Per-token probes for the incongruent condition, Llama-3.2-3B, level 3, CoT. Top: how often the best-layer probe reads the true value (green) or the name-suggested value (red). Bottom: true-value accuracy by layer. The displayed tokens are from one example; the rates pool problems with different values and names. Dashed line: start of the supplied correct chain; dotted line: position before the target value; arrow: the supplied value token. Shading marks the name’s tokens.

Model	Level	Regime	Neutral accuracy (%)	number word on the queried variable		number word on the intermediate variable	
				helps	errors	helps	errors
Llama-3.1-8B	1	CoT	99 [98,100]	+0.0	+0.0	+0.3	+0.0
Llama-3.1-8B	1	direct	74 [38,99]	+16.1*	+11.2*	+8.8*	+1.3
Llama-3.2-3B	1	CoT	97 [95,100]	+1.1	+0.1	-2.4	+0.3
Llama-3.2-3B	1	direct	46 [21,93]	+0.6	+5.9*	-3.9*	+2.6*
Llama-3.1-8B	2	CoT	100 [100,100]	-0.1	+0.0	-0.0	+0.0
Llama-3.1-8B	2	direct	36 [33,37]	+11.8*	+0.7	+2.6	+2.2
Llama-3.2-3B	2	CoT	94 [90,98]	+3.7	+0.5*	+1.0*	-0.1
Llama-3.2-3B	2	direct	29 [27,30]	+10.8*	+2.5*	+2.6*	+2.2
Llama-3.1-8B	4	CoT	88 [63,100]	+1.0	-0.0	-2.3	+0.2
Llama-3.1-8B	4	direct	28 [26,31]	+13.7*	+5.8	+6.0*	+0.2
Llama-3.2-3B	4	CoT	64 [26,85]	-10.2*	+0.7	-2.6*	+0.4
Llama-3.2-3B	4	direct	20 [16,23]	-1.6	+1.9	-0.2	+1.2*
Llama-3.1-8B	5	CoT	65 [43,100]	+1.2	-0.3	+0.1	-0.1
Llama-3.1-8B	5	direct	20 [13,26]	+20.2*	+20.5*	+0.9	+0.4
Llama-3.2-3B	5	CoT	69 [48,97]	-1.4	+0.1	-2.5	-0.1
Llama-3.2-3B	5	direct	12 [11,12]	+4.7*	+3.5*	+1.1*	+0.5

Table 9: As Table 1, for levels 1, 2, 4 and 5 (Llama-3.2-3B and Llama-3.1-8B). Level 1 is the no-computation control: the intermediate variable’s value is stated rather than computed.

rates, and the error taxonomy behind the word-class effect.

J Additional value readouts

We extend the value-step comparison to Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-12B-Instruct. Each probe trains on 10,000 separate neutral problems with three probe seeds, using demonstration set 7. Table 10 reports the added pairs; the main comparison includes them. We select layers by neutral selectivity, as in the original analysis. The misleading-name

accuracy uses the same selected layer. This extension tests the aggregate association; it does not resolve the within-problem link.

We also probe every token for Gemma-3-4B-Instruct (Figures 8 and 9), using 4,000 independent neutral training problems and 1,000 test problems per condition. The tokens shown come from one example; the curves and heatmaps pool problems with different names and values. Maximum-over-layers curves describe where a digit can be read; they do not identify the representation used to answer. On neutral problems, the best pre-chain accuracy is 26.1% for the queried digit and 50.3% for the intermediate digit. The queried readout first exceeds 90% at token 24; the intermediate readout first exceeds 90% at token 13. Token zero is the first token of the chain; negative positions are in the prompt.

Model	Variable	Layer	Neutral	Control	Misleading
Llama-3.2-3B-Instruct	queried	18	99.9	26.3	100.0
Llama-3.2-3B-Instruct	intermediate	19	99.9	29.5	99.9
Llama-3.1-8B-Instruct	queried	20	100.0	21.0	100.0
Llama-3.1-8B-Instruct	intermediate	20	100.0	25.7	100.0
Gemma-3-12B-it	queried	40	100.0	13.1	100.0
Gemma-3-12B-it	intermediate	42	100.0	11.9	100.0

Table 10: Additional probes at each variable’s value-writing position. Neutral and misleading columns are true-digit accuracy; control is neutral name-identity accuracy. Rates are percentages, averaged over three probe seeds. The layer maximizes neutral accuracy minus control accuracy.

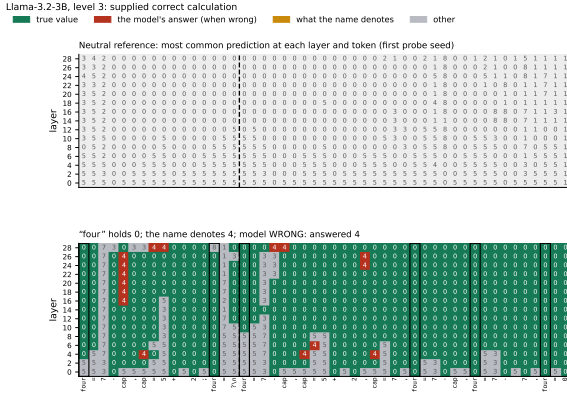


Figure 6: Adapted from Kudo et al.’s Figure 3. Read columns left to right as tokens are processed; rows run from early model layers at the bottom to later layers at the top. Each cell contains a separate predictor’s digit, not a generated digit. Probes read a supplied correct calculation for a problem the model answers with the name-suggested digit. Bottom: majority prediction over three probe seeds at each layer and token; green the variable’s true value, red the value the model answered, amber the value its name denotes, grey anything else. Top: the most common prediction on neutral problems from the first probe seed, for reference. Boxes mark name tokens; the dashed line marks the calculation’s start. Matching the neutral reference alone does not establish a cell’s information content.

Top-1 probe prediction over the model's own chain (Llama-3.2-3B, L3, with chain of thought; probes trained at the value step of "v1" on neutral problems; 8 instances)

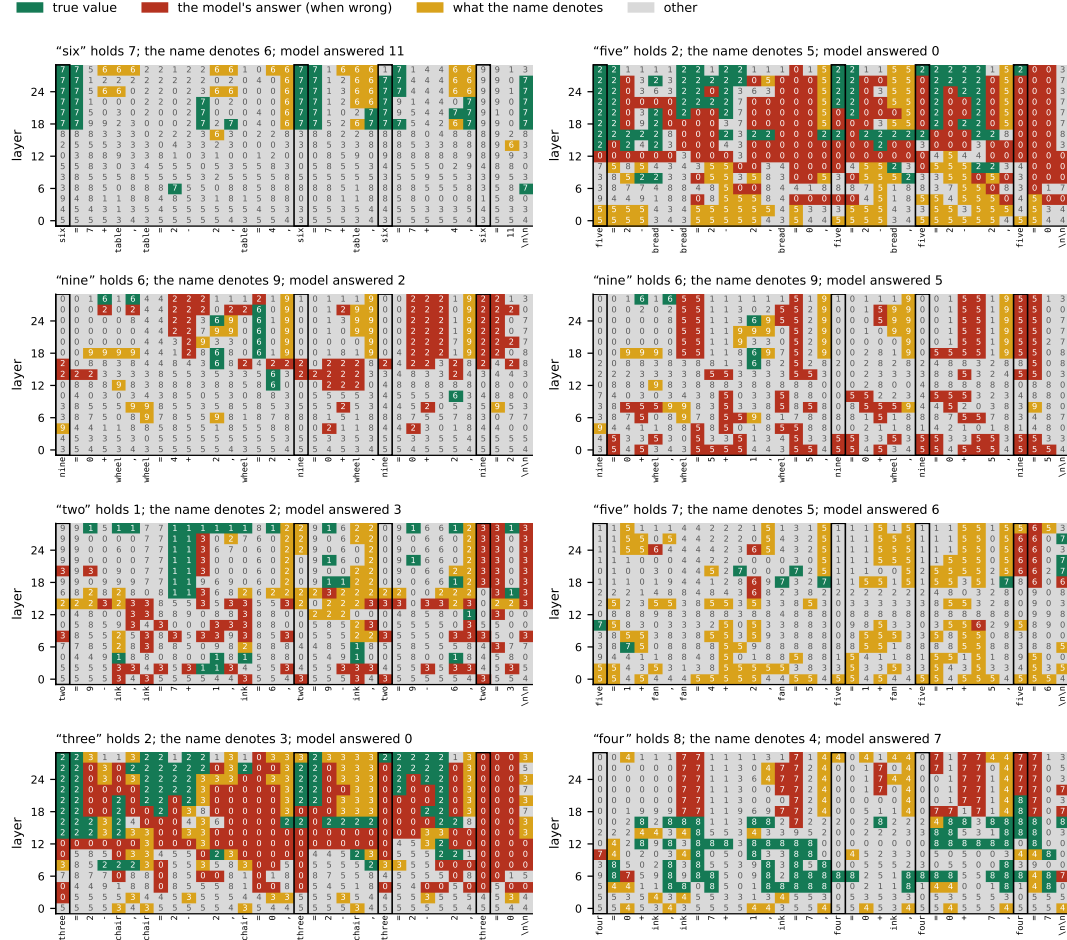


Figure 7: Probes trained at the value step, applied to the model's own generated chain rather than to a force-decoded gold chain, for the 8 level-3 incongruent problems that Llama-3.2-3B answered wrongly under CoT (the first 8 of 108 wrong answers in 2,000 instances). Probes are trained on neutral problems at the step that writes the variable's value and applied at every generated token and every second layer; colours as in Figure 6, boxed columns the name's own tokens. This small sample contains no name errors and is illustrative rather than an estimate of their rate. The fixed-sample transfer analysis and separate additional name-error cases appear in Appendix M.

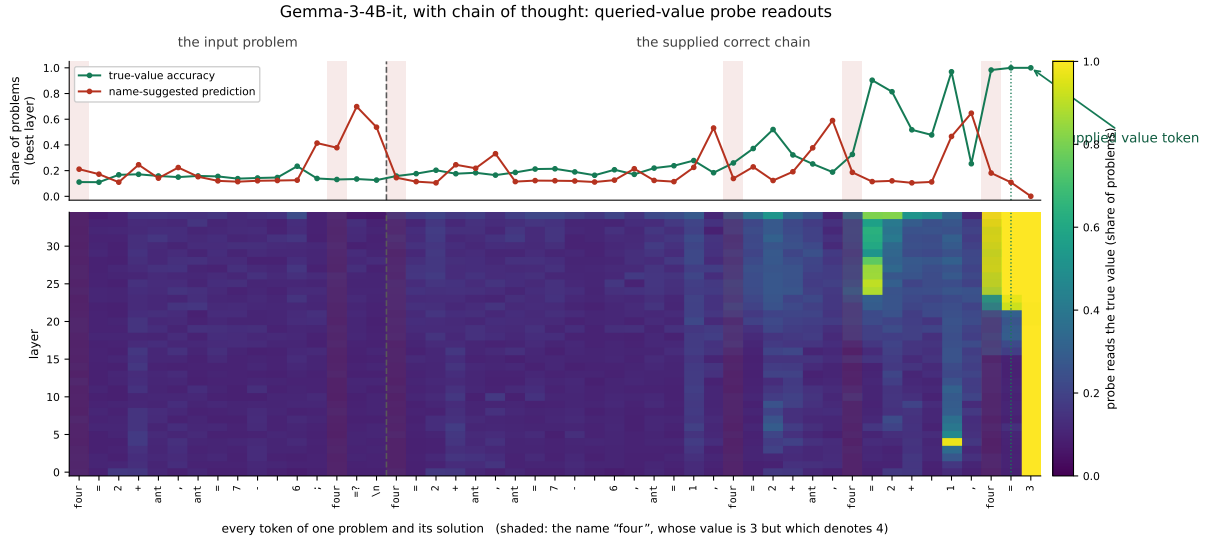


Figure 8: Gemma-3-4B-Instruct, queried variable, misleading-name problems under the correct chain. Top: maximum-over-layers true-value accuracy and name-suggested prediction rate; bottom: true-value accuracy by layer. Shading marks the variable's name; the dotted line marks the position before its value; the arrow marks the supplied digit itself.

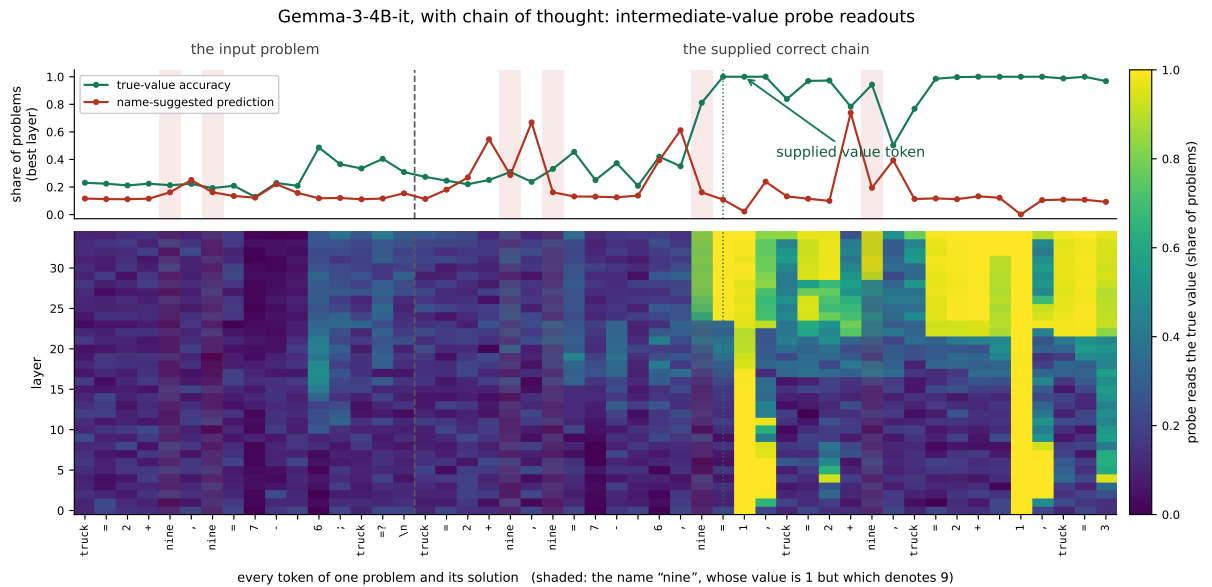


Figure 9: The same Gemma token sweep for the intermediate variable. Both figures use neutral-trained probes; no misleading-name problems enter training.

K Every figure at full size

Each figure below is reproduced as a full-width figure on a portrait page, within the ACL margins.

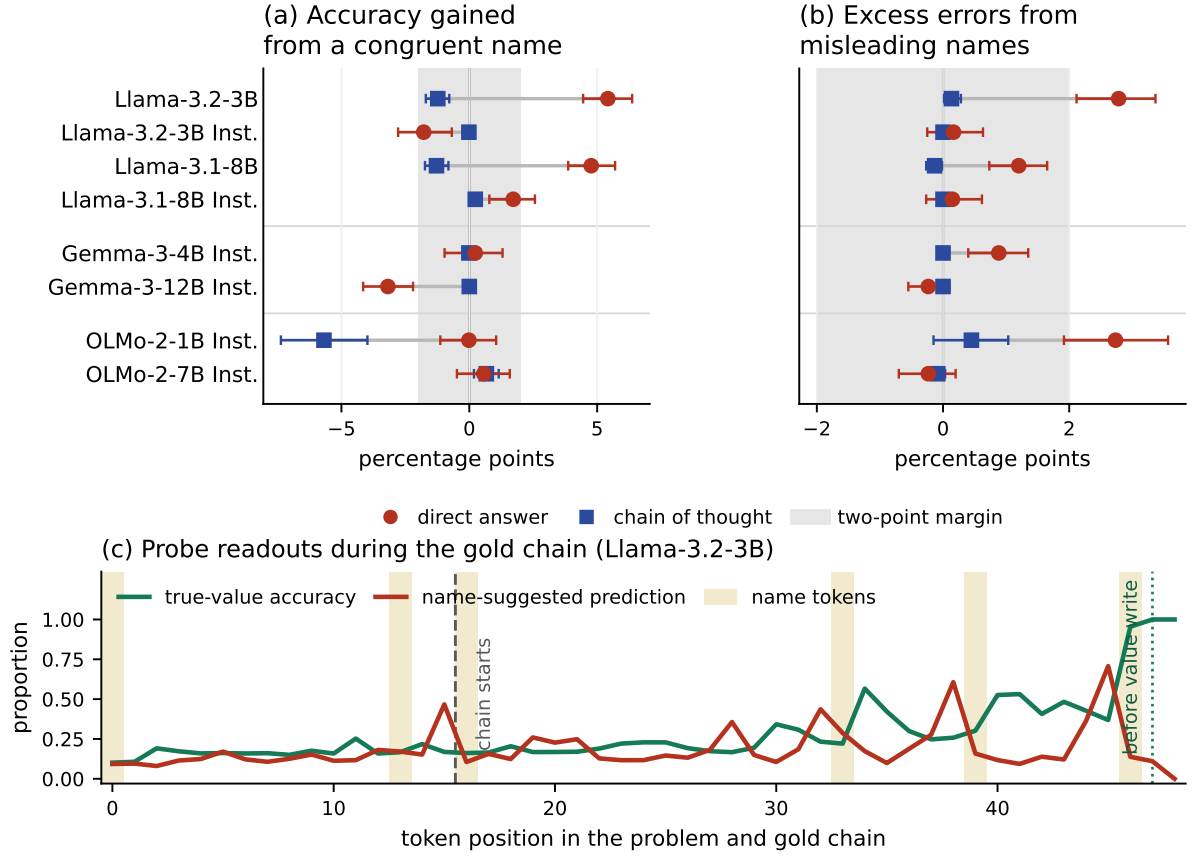


Figure 10: Two-operation arithmetic tasks, queried variable. Left: the accuracy gain from a name denoting the correct digit. Middle: added answers matching a misleading digit relative to the ordinary-name twin. Red circles request only an answer; blue squares request calculations first. Points average three demonstration sets; bars are 95% intervals. Grey bands mark two points, not a 90% equivalence test. Right: a separate predictor reads the correct digit (green) or name-suggested digit (red) as the correct calculation is supplied to Llama-3.2-3B. These are best-layer readouts, not generated outputs. OLMo-2-1B is below the planned neutral-calculation accuracy floor.

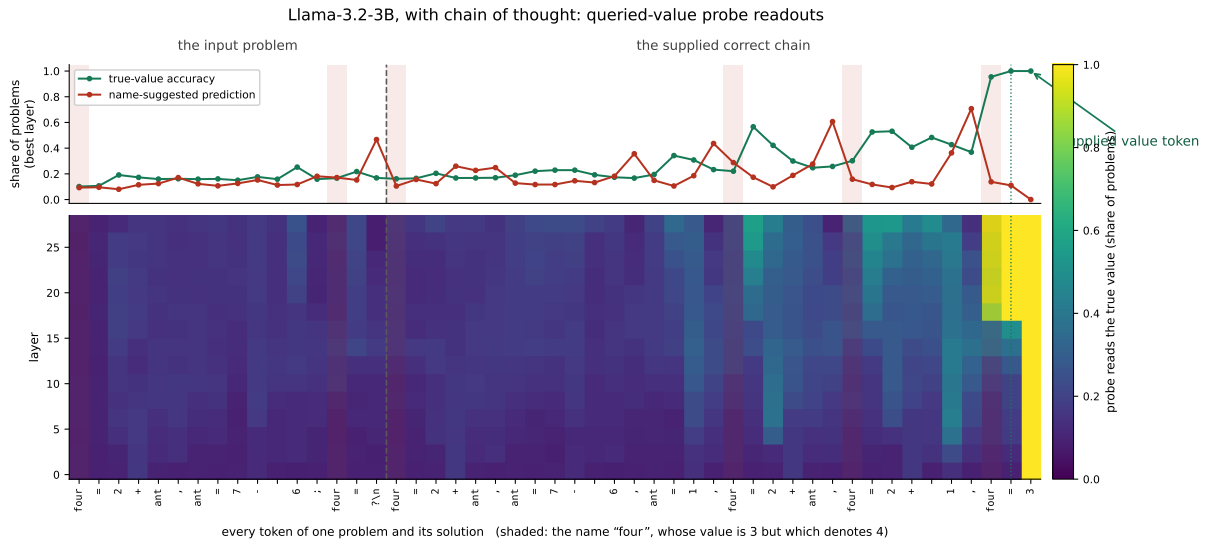


Figure 12: Per-token probes, chain-of-thought regime, queried variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

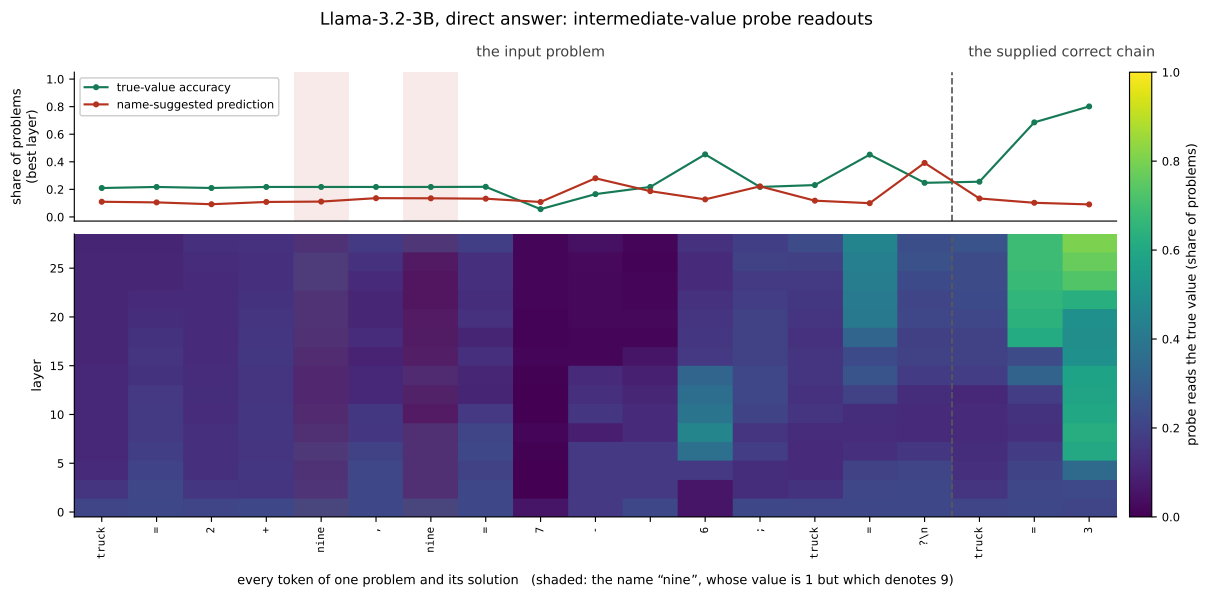
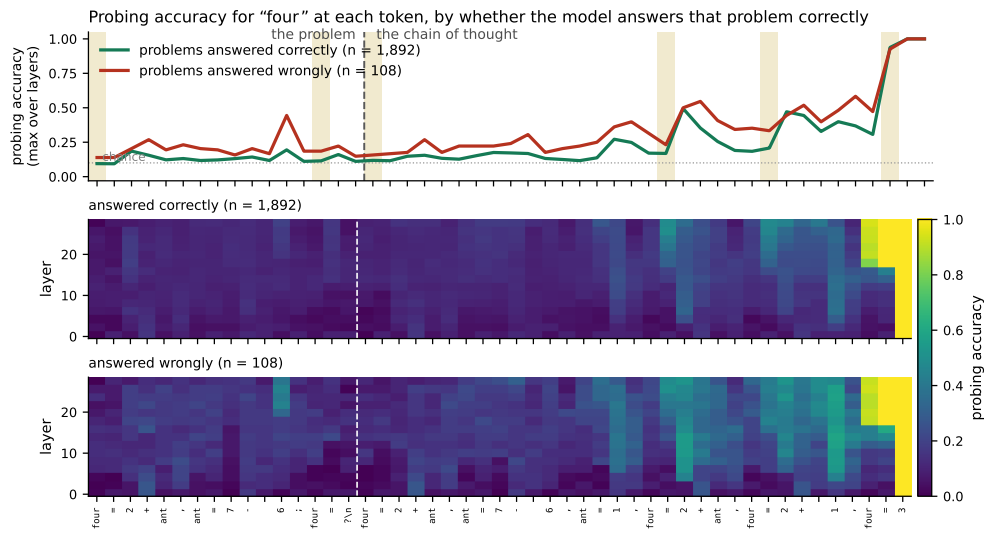


Figure 13: Per-token probes, direct regime, intermediate variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.



Probes read the variable's true value; states come from a forced pass over the correct chain, so the lower panel asks what the state holds for problems this model gets wrong unaided.

Figure 14: Probing accuracy split by whether the model answers the problem correctly on its own, with the layer sweep for each half. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

Llama-3.2-3B, level 3: supplied correct calculation

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

true value
 the model's answer (when wrong)
 what the name denotes
 other

Neutral reference: most common prediction at each layer and token (first probe seed)

[illegible]

"four" holds 0; the name denotes 4; model WRONG: answered 4

[illegible]

Figure 15: Top-1 probe prediction at every layer and token, on a problem answered with the name-suggested value. Layout after [Kudo et al.](#)’s Figure 3. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3: supplied correct calculation

■ true value ■ the model's answer (when wrong) ■ what the name denotes ■ other

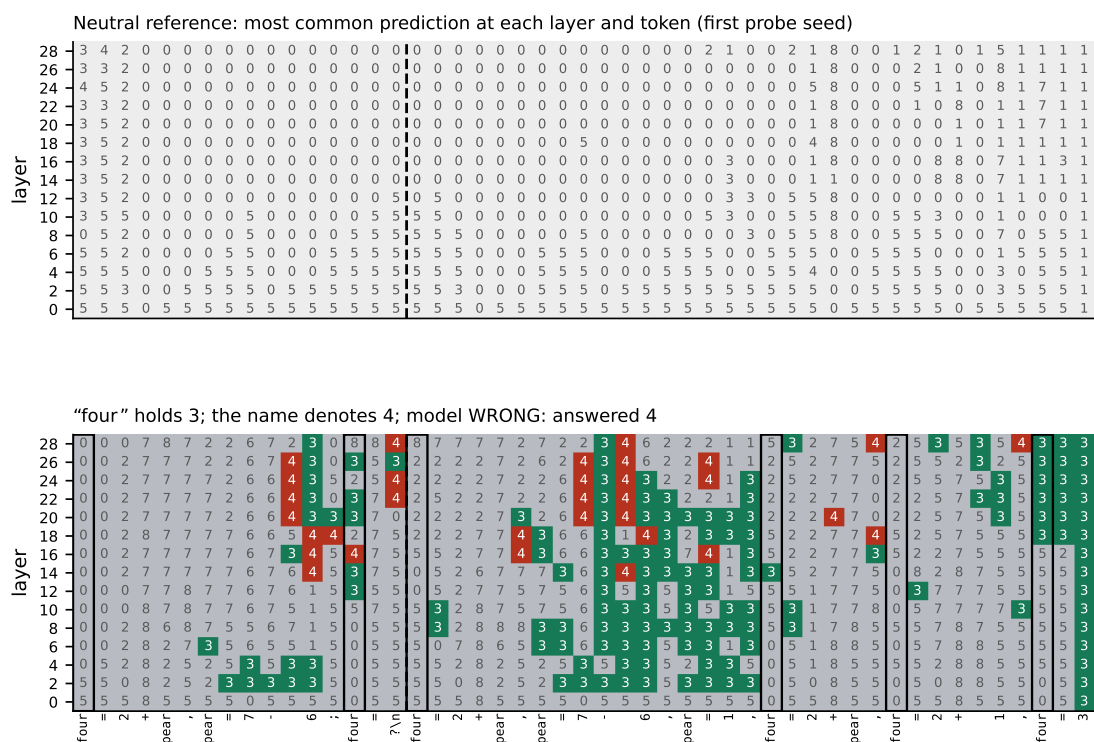


Figure 16: Queried-variable example answered with the digit suggested by its misleading name. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3: supplied correct calculation

■ true value ■ the model's answer (when wrong) ■ what the name denotes ■ other

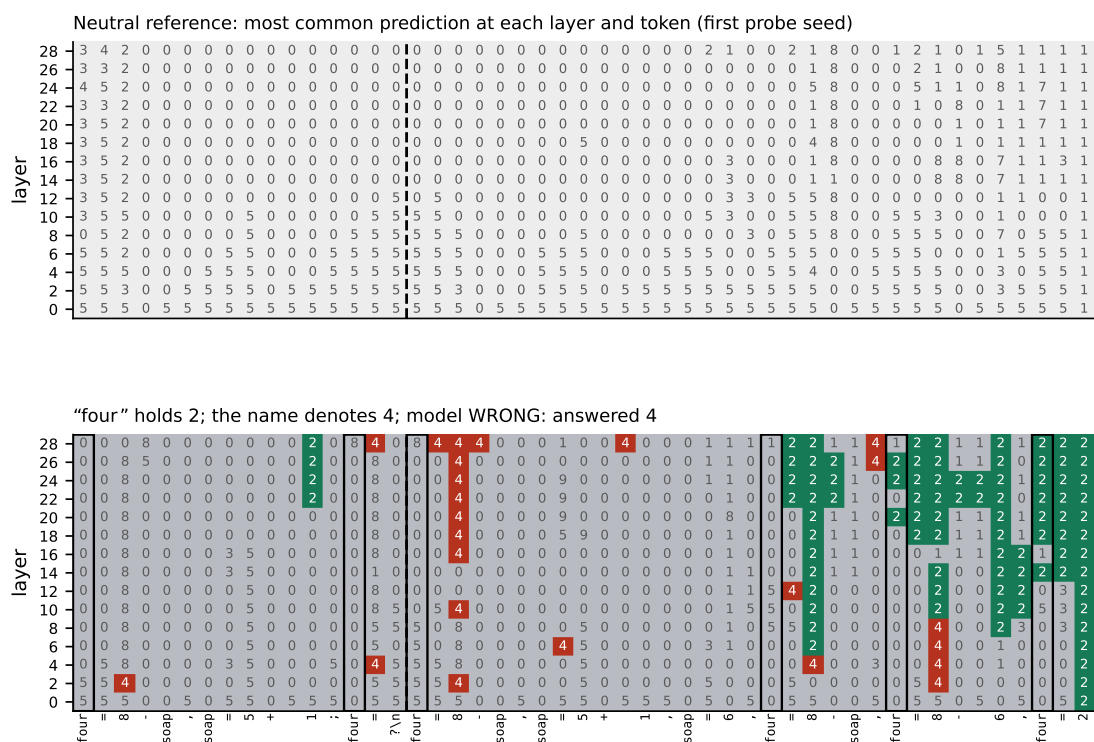


Figure 17: Queried-variable example answered with the digit suggested by its misleading name. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3: supplied correct calculation

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

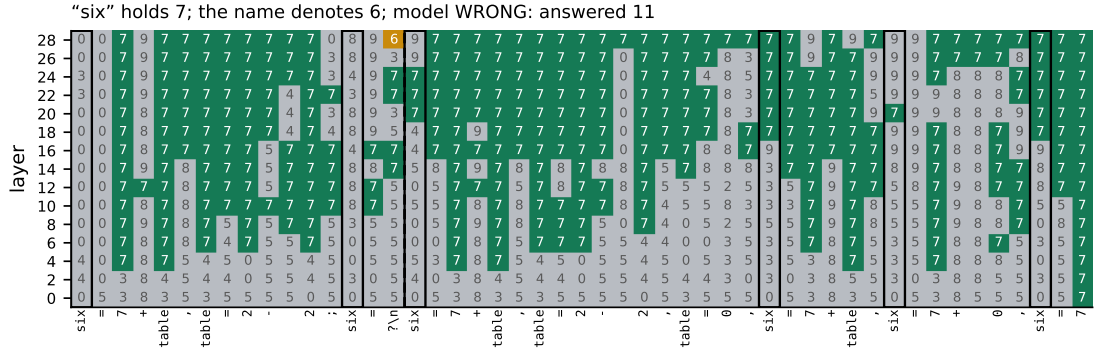
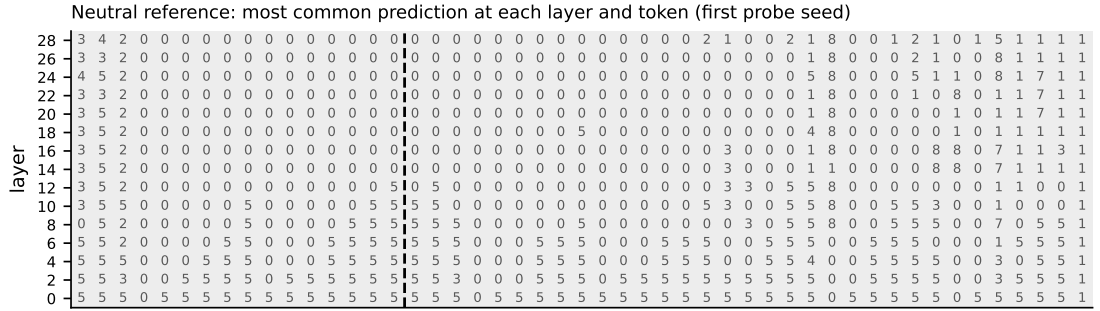


Figure 18: Queried-variable example whose generated final answer is wrong and differs from the name-suggested digit. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3: supplied correct calculation

■ true value ■ the model's answer (when wrong) ■ what the name denotes ■ other

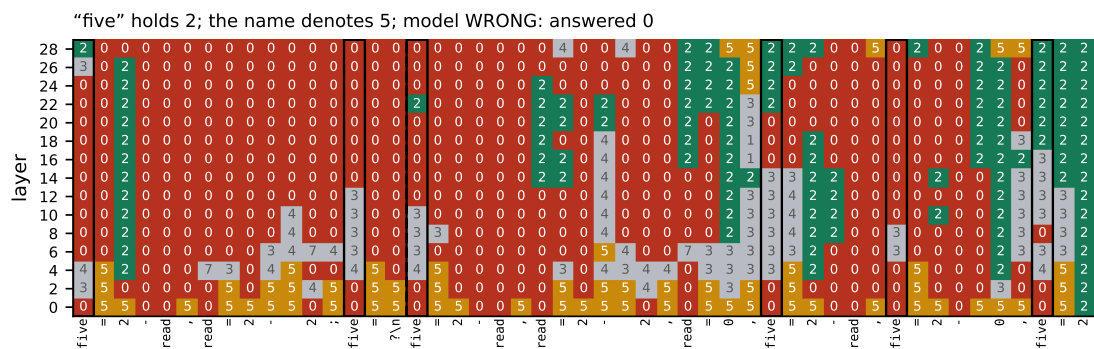
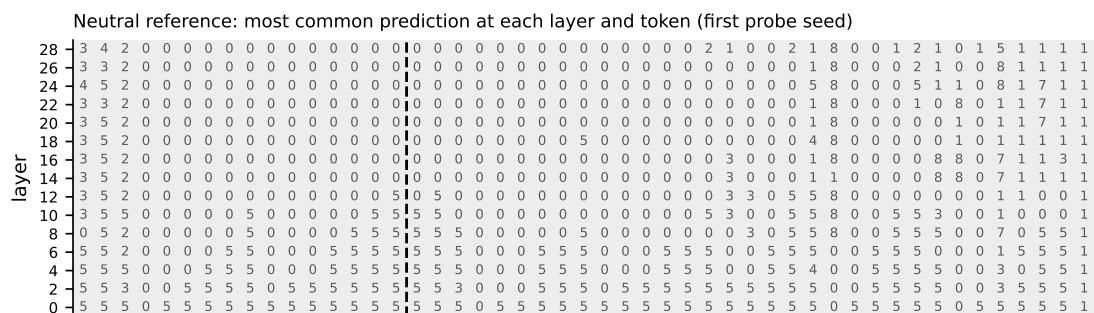


Figure 19: Queried-variable example whose generated final answer is wrong and differs from the name-suggested digit. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3: supplied correct calculation

true value
 the model's answer (when wrong)
 what the name denotes
 other

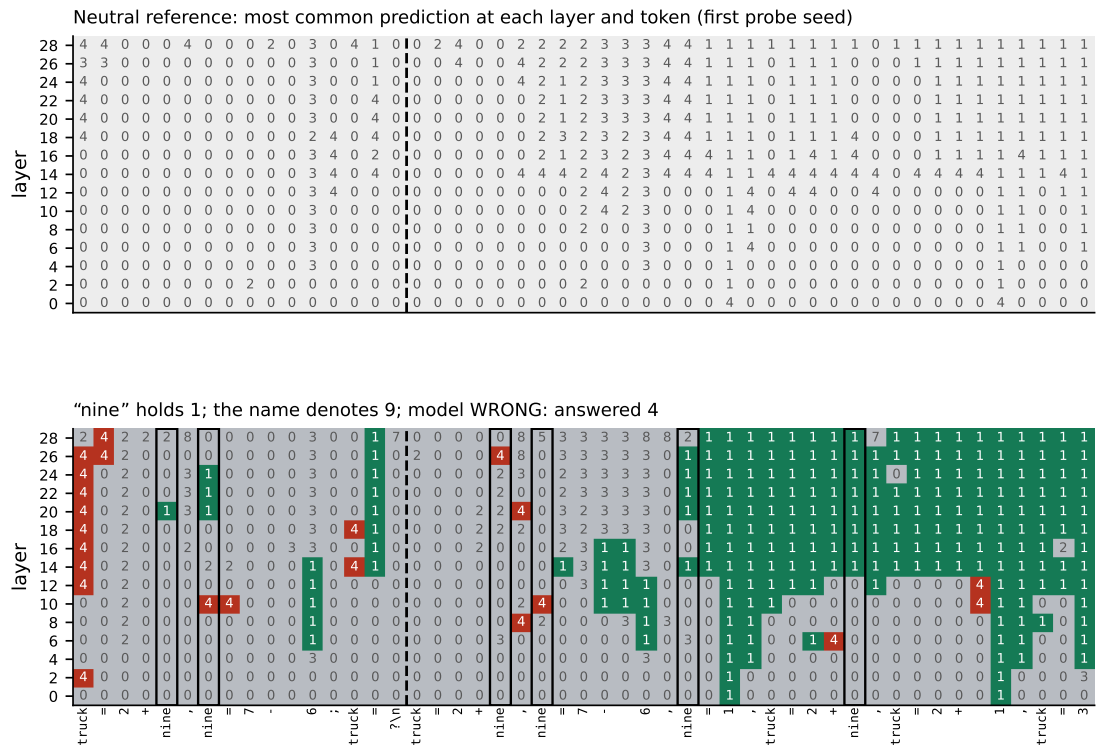


Figure 20: Intermediate-variable example from a problem answered wrongly. Red is the model’s final answer, which is the queried variable’s value, so a cell that matches it is not a readout of this variable. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3: supplied correct calculation

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

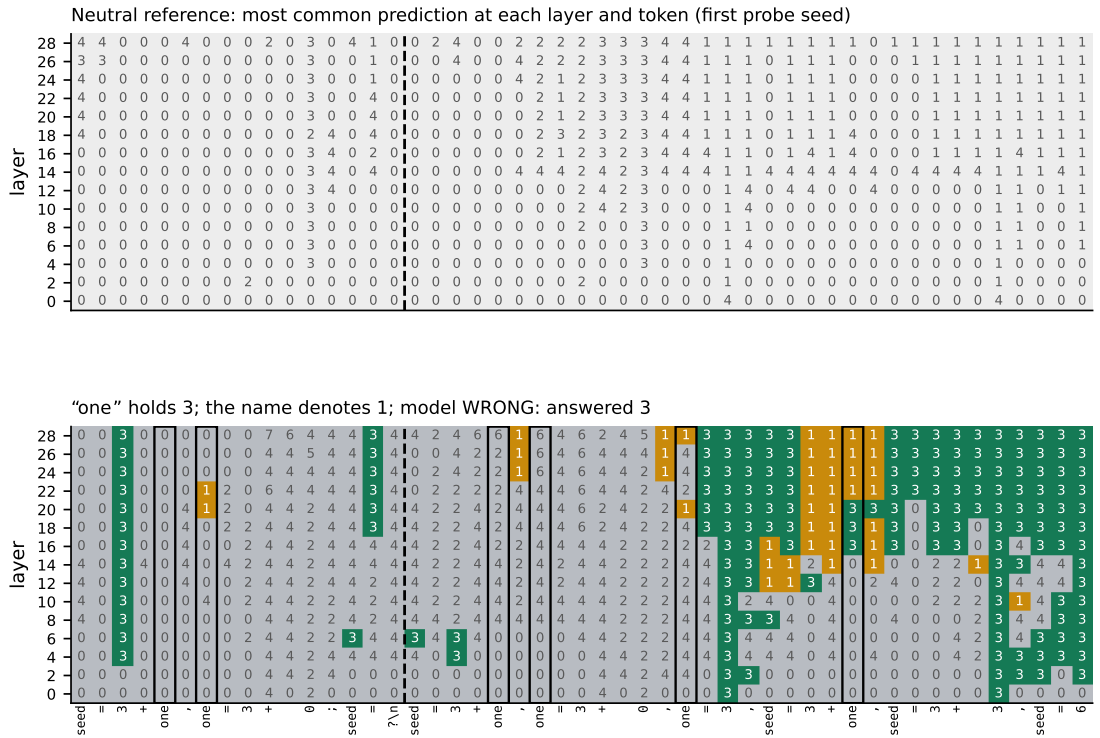


Figure 21: Intermediate-variable example from a problem answered wrongly. Each column is a token of the displayed problem or supplied correct calculation; each row is a model layer, from early (bottom) to late (top). Each digit is the predictor’s most common output over three training seeds. Green matches the variable’s correct value; red matches the model’s wrong final answer; amber matches the name’s suggestion; grey is another digit. The top strip is the neutral condition’s most common prediction from one seed. It is a reference, not an information test. These predictions read supplied correct calculations, not the model’s generated mistake.

Llama-3.2-3B, level 3, direct answer: per-token probes for the value of the intermediate variable (trained on neutral problems)

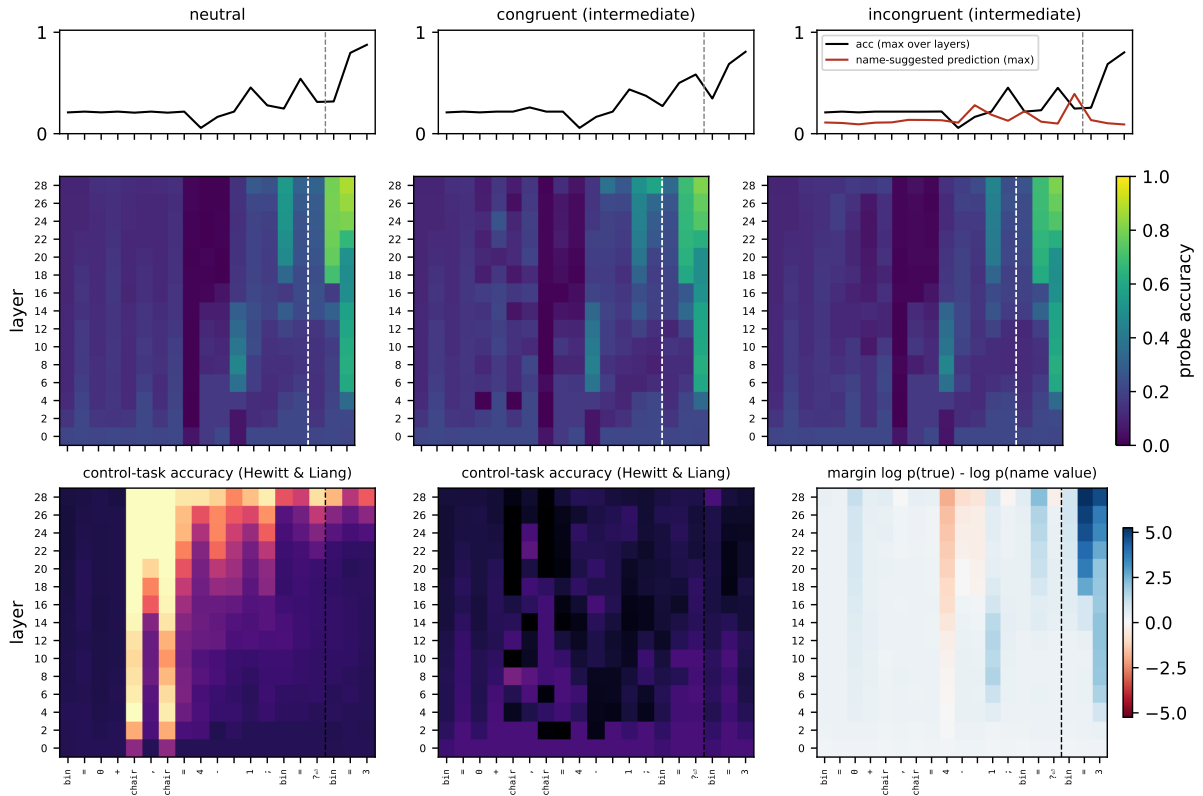


Figure 23: The same heatmaps in the direct regime. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

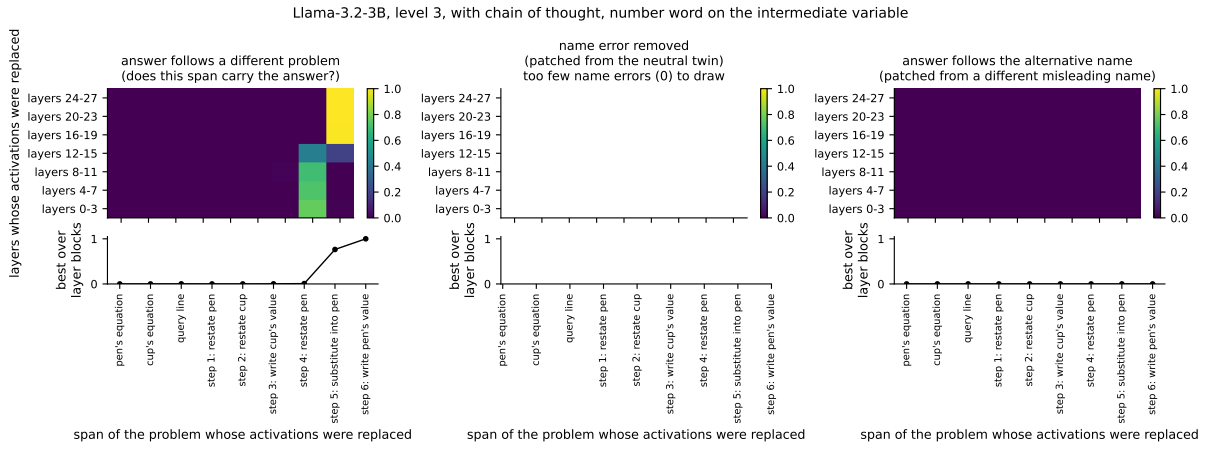


Figure 24: Span-by-layer-window patching grids, three sources (the error-removal panel is empty: with CoT there are too few name errors to draw), after [Kudo et al.](#)’s Figures 5–6. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

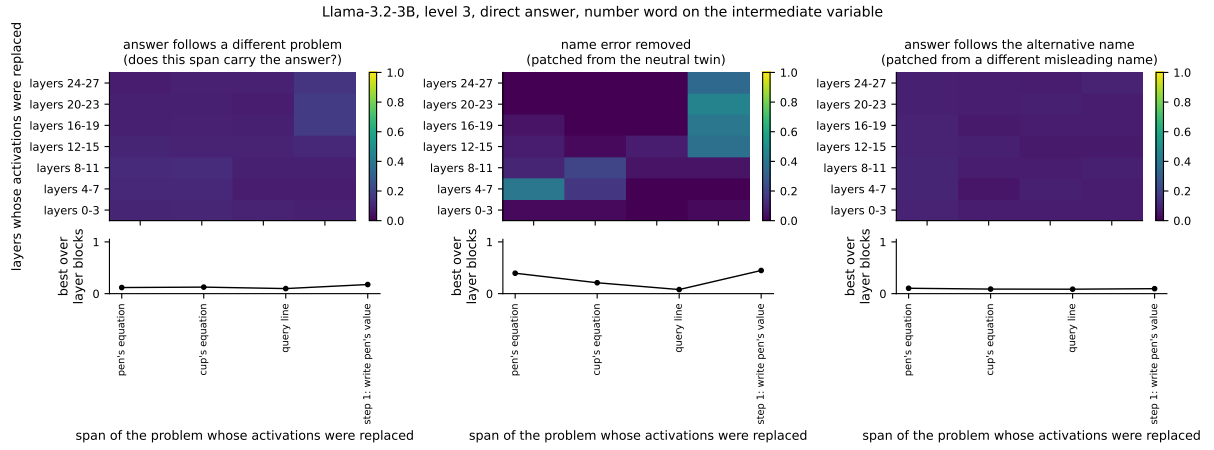


Figure 25: The same grids in the direct regime. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

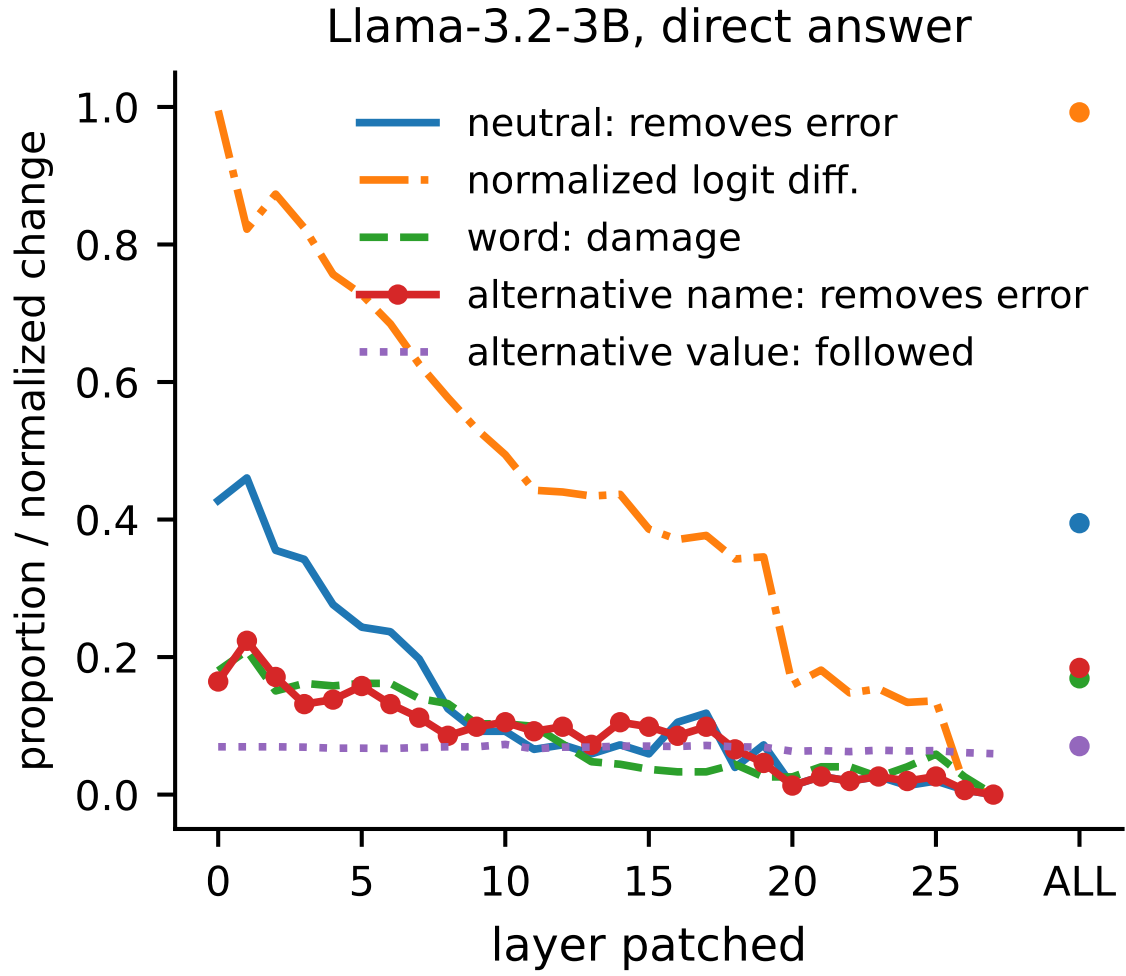


Figure 26: Name errors removed by patching, by layer, with both controls. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

Llama-3.2-3B, level 3, with chain of thought: per-token probes for the value of the queried variable (trained on neutral problems)

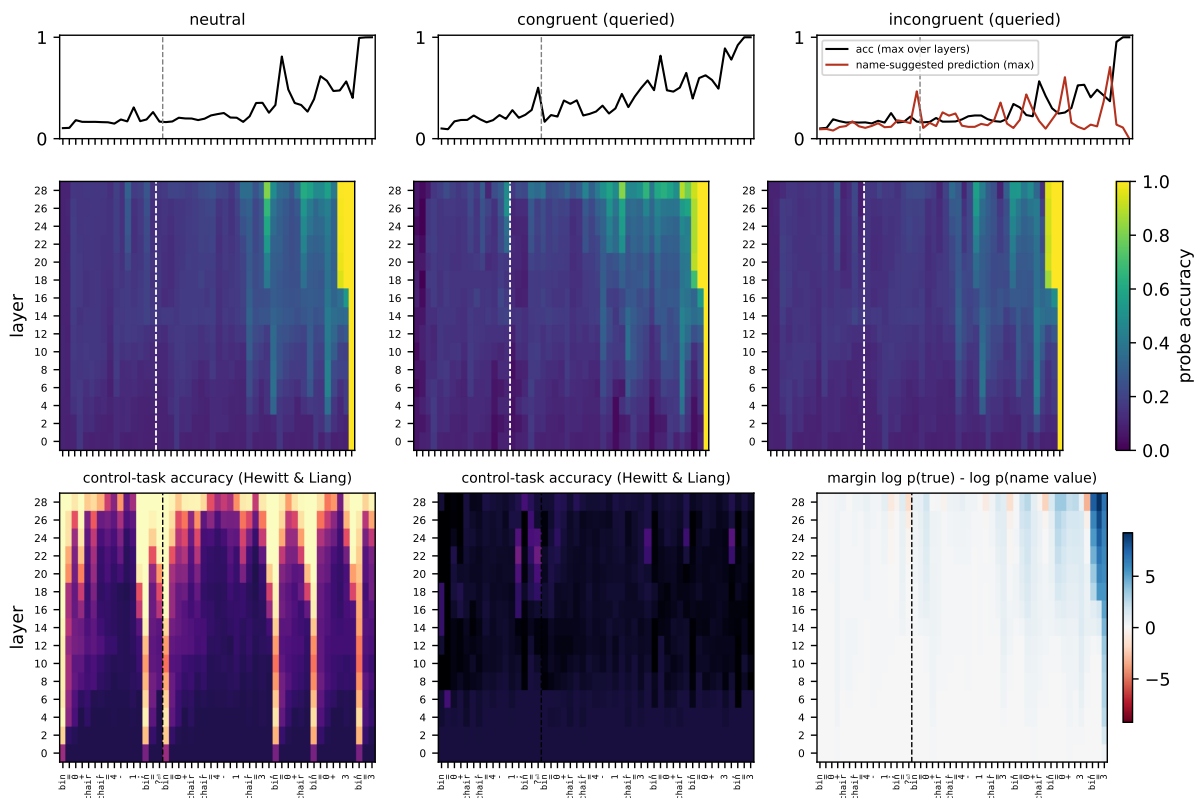


Figure 27: Accuracy heatmaps as above, chain-of-thought regime, queried variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

Llama-3.2-3B, level 3, direct answer: per-token probes for the value of the queried variable (trained on neutral problems)

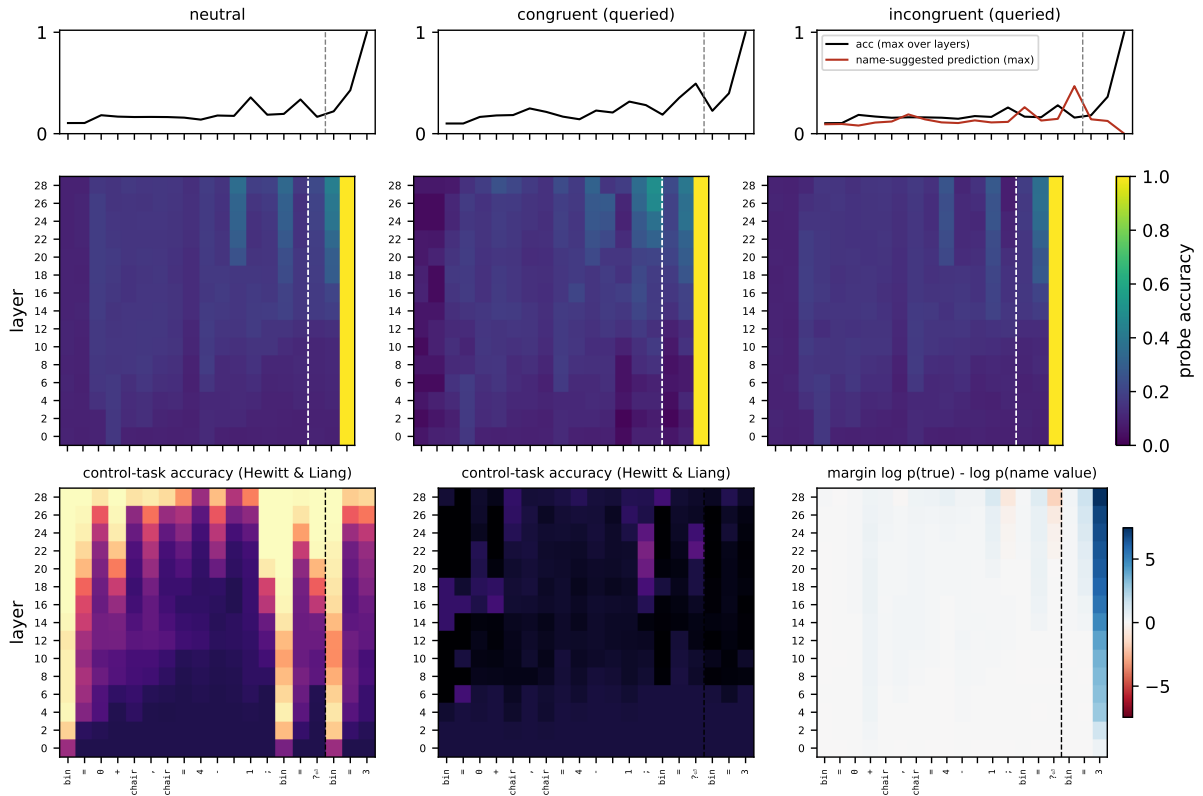


Figure 28: The same heatmaps in the direct regime, queried variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

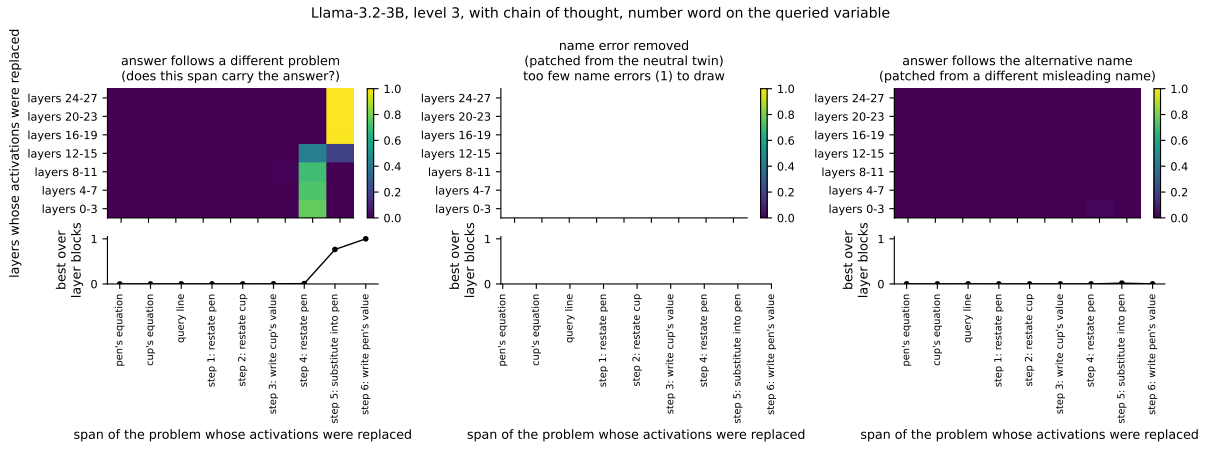


Figure 29: Span-by-layer-window patching grids, chain-of-thought regime, queried variable; the error-removal panel is empty for the same reason. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

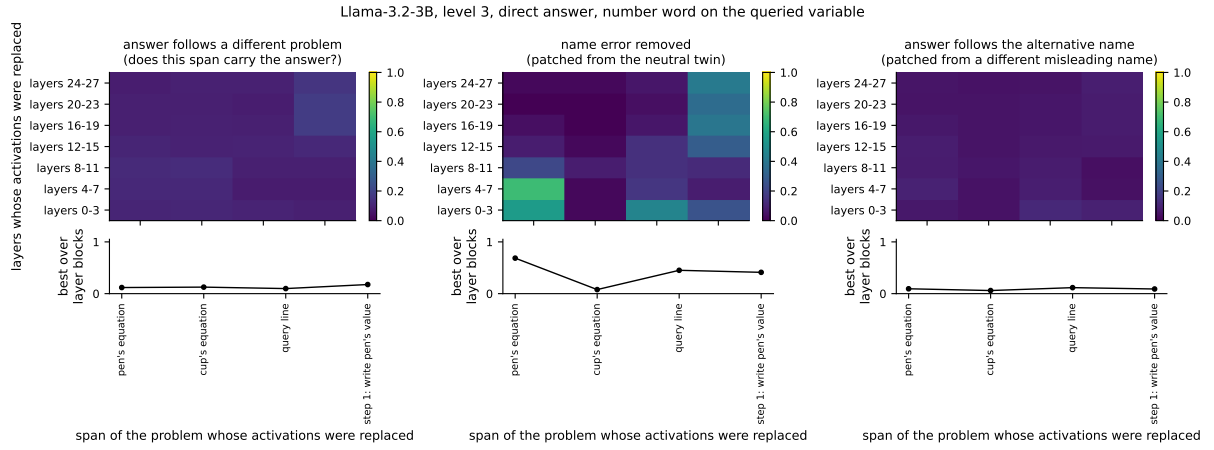


Figure 30: The same grids in the direct regime, queried variable. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

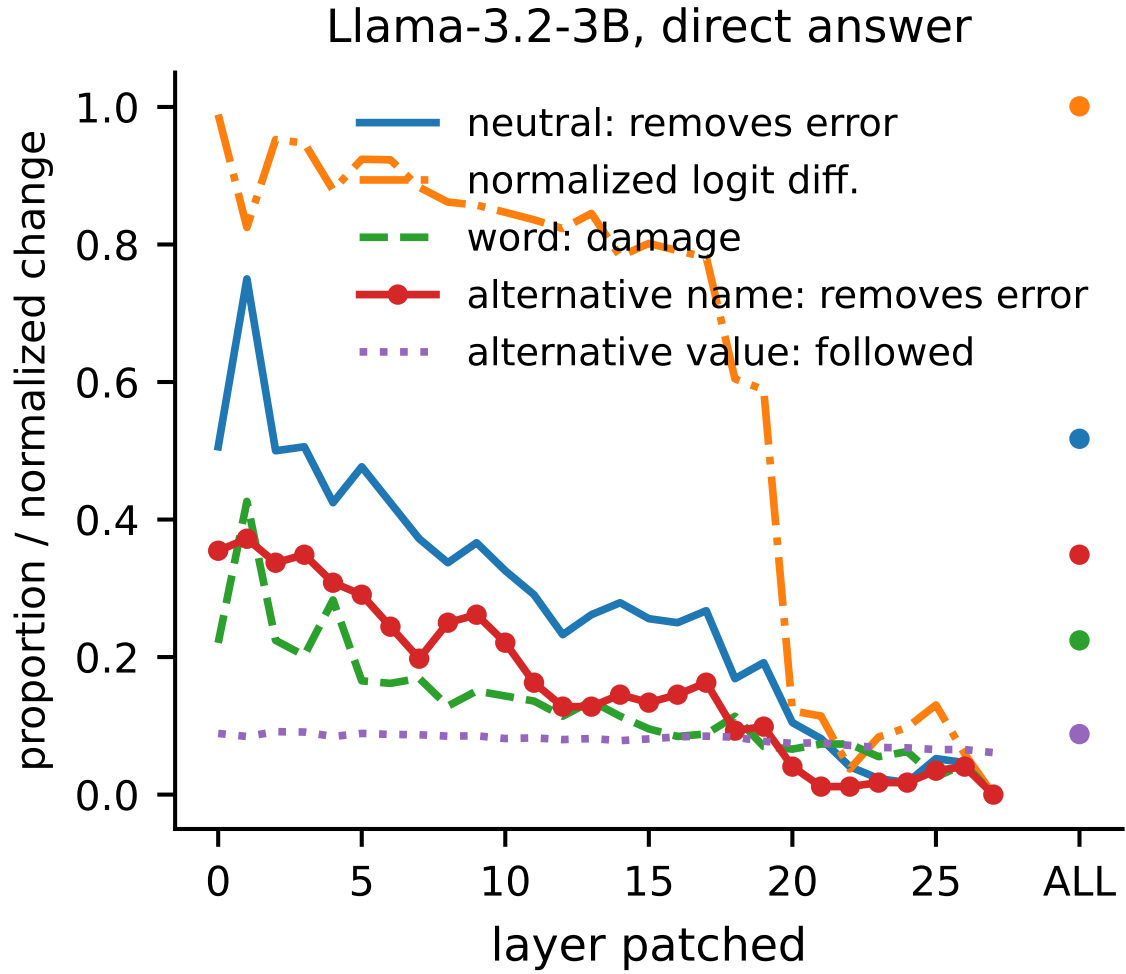


Figure 31: Name errors removed by patching, by layer, with both controls, queried variable. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

Top-1 probe prediction over the model's own chain (Llama-3.2-3B, L3, with chain of thought; probes trained at the value step of "v1" on neutral problems; 8 instances)

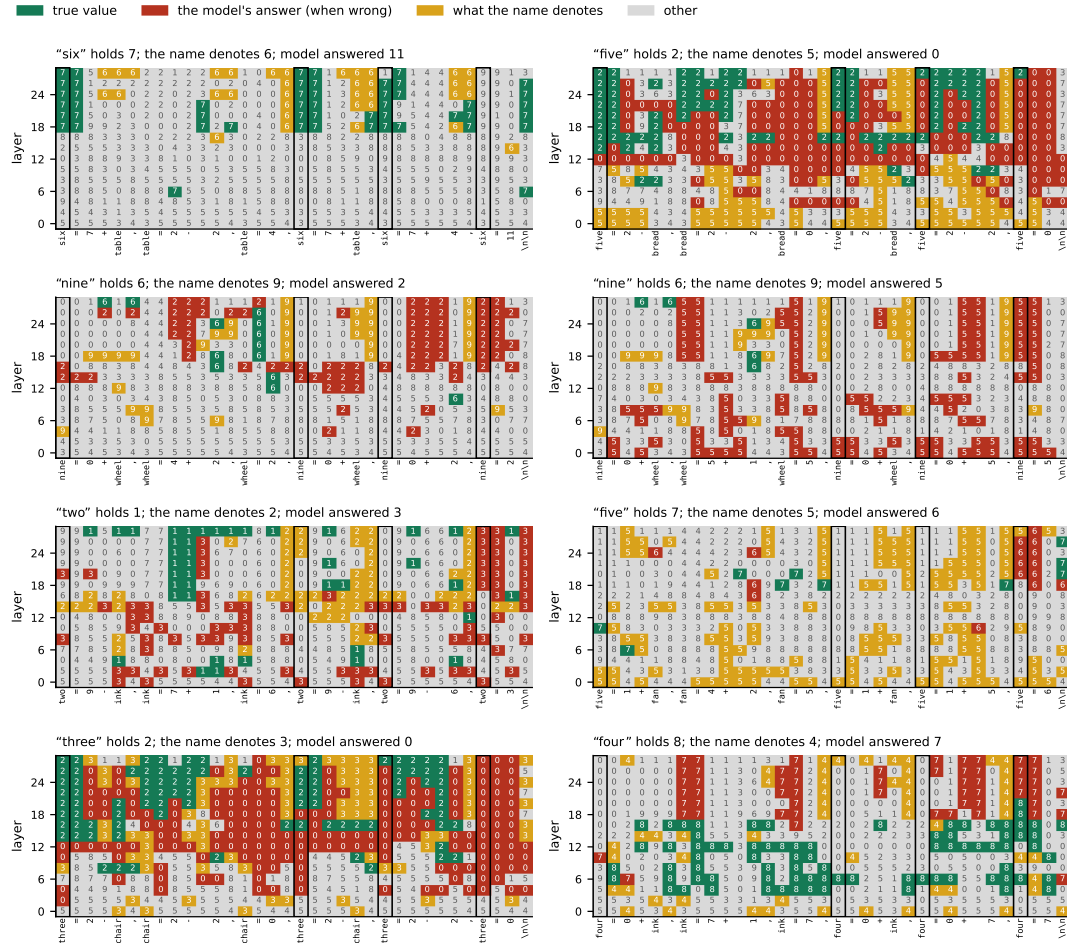


Figure 32: Llama-3.2-3B, queried variable: internal predictions during eight calculations the model generated before giving a wrong final answer. Here the columns are tokens the model actually generated. Rows are model layers and cell digits are separate predictor outputs. Green matches the correct variable value, red the wrong final answer, amber the name's suggestion, and grey another digit. Boxes mark name tokens. Each panel is one problem; these examples do not estimate an error rate.

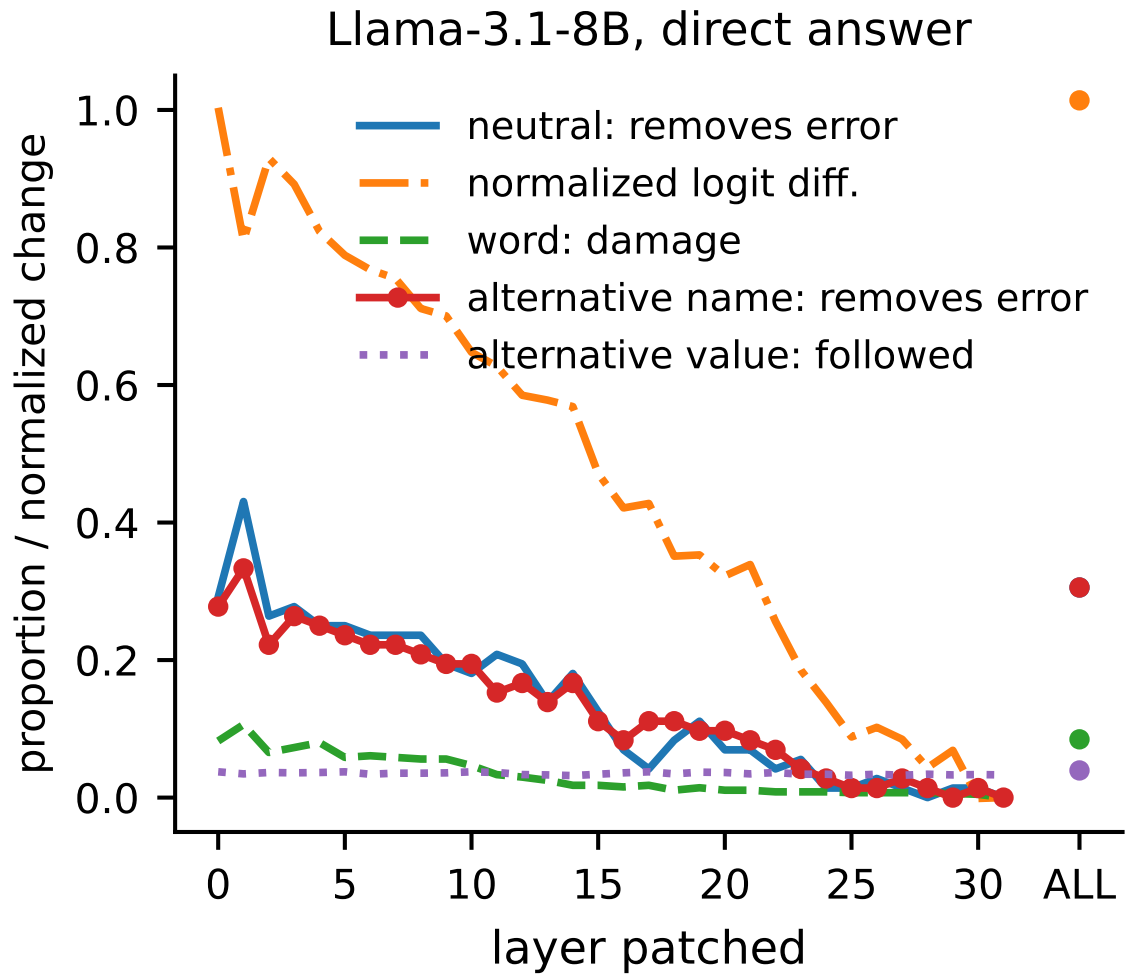


Figure 33: Llama-3.1-8B, name errors removed by patching, by layer, with both controls, direct regime, intermediate variable. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

Llama-3.1-8B, level 3, with chain of thought: per-token probes for the value of the queried variable (trained on neutral problems)

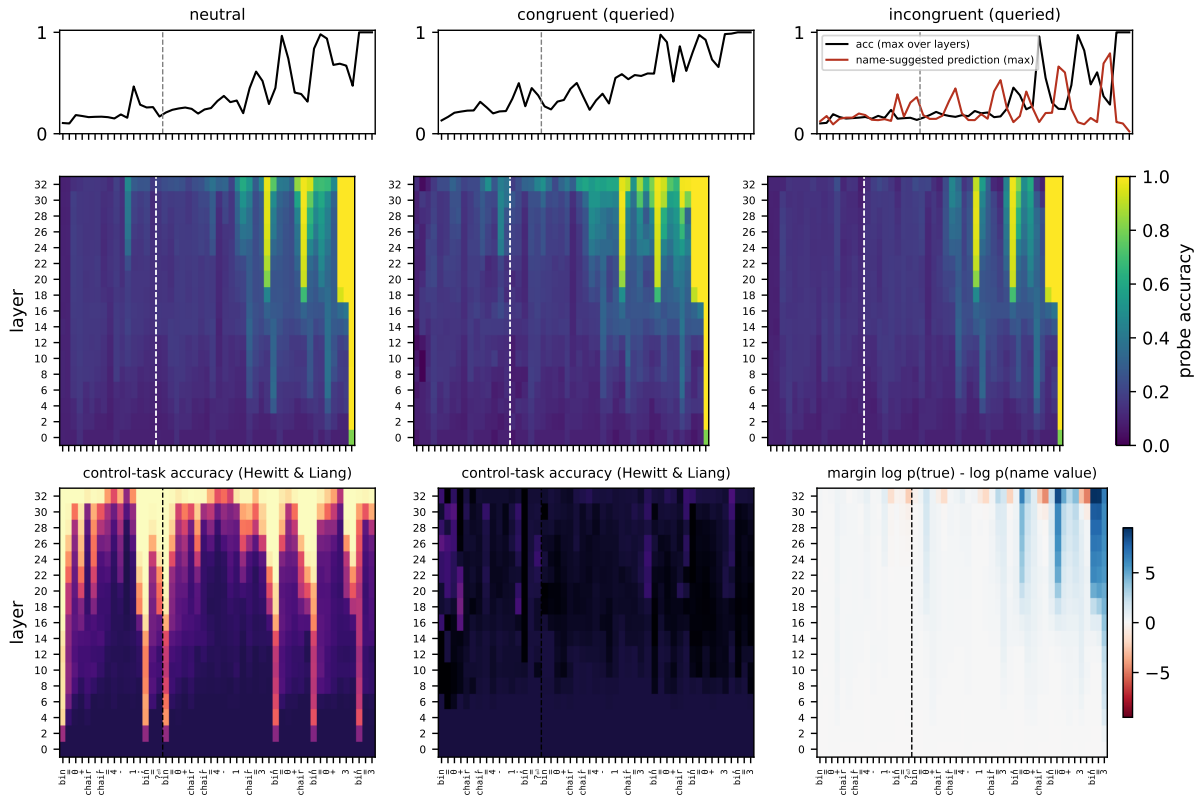


Figure 34: Llama-3.1-8B, chain-of-thought regime, queried variable: the accuracy heatmaps of the Llama-3.2-3B figures. The value is decodable from the equation that restates the variable’s definition, before the chain writes it. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

Llama-3.1-8B, level 3, with chain of thought: per-token probes for the value of the intermediate variable (trained on neutral problems)

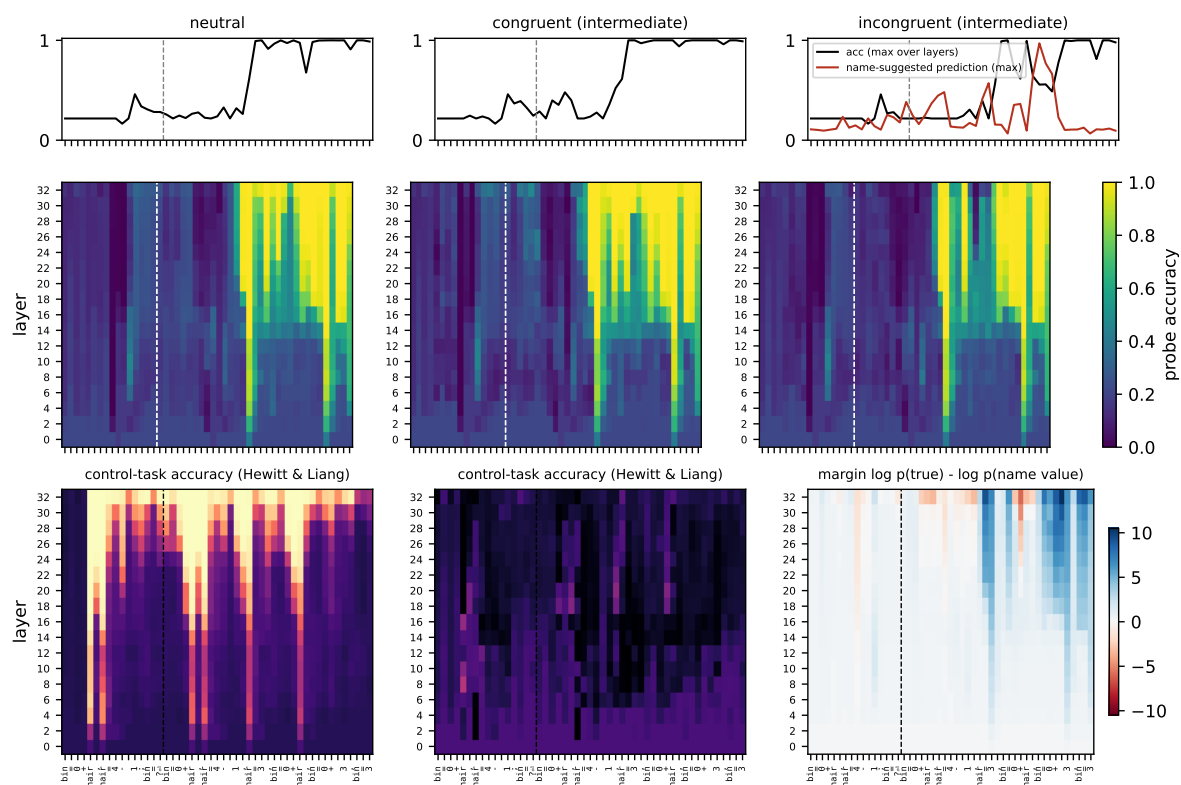


Figure 35: Llama-3.1-8B, chain-of-thought regime, intermediate variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

Llama-3.1-8B, level 3, direct answer: per-token probes for the value of the intermediate variable (trained on neutral problems)

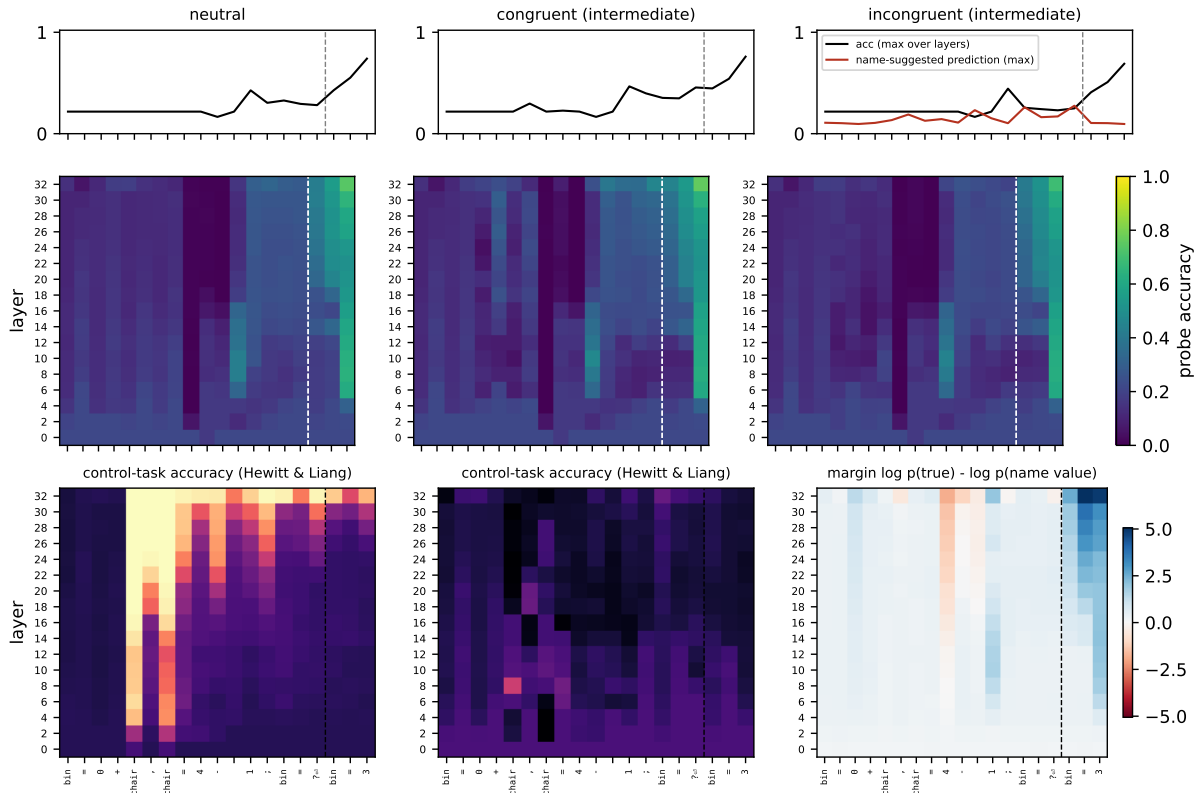


Figure 36: Llama-3.1-8B, direct regime, intermediate variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

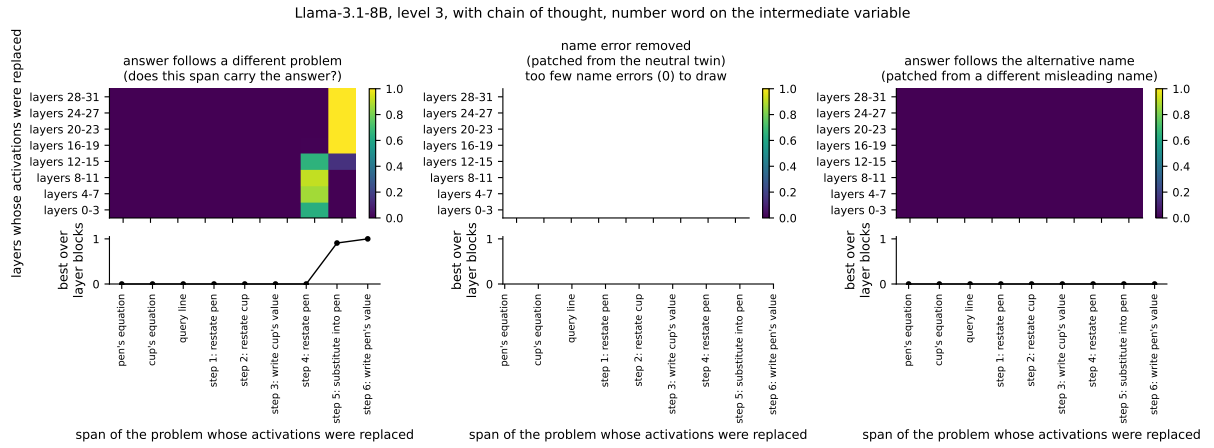


Figure 37: Llama-3.1-8B, span-by-layer-window patching grids, chain-of-thought regime. The error-removal panel is empty: none of the 500 patched problems is answered with the name-suggested value. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

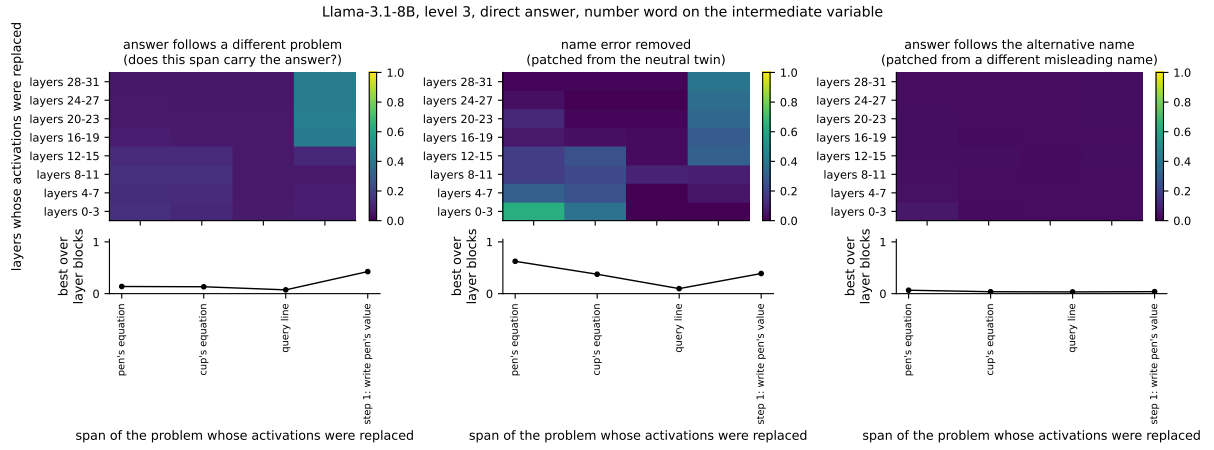


Figure 38: Llama-3.1-8B, the same grids in the direct regime. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

Llama-3.1-8B, level 3, direct answer: per-token probes for the value of the queried variable (trained on neutral problems)

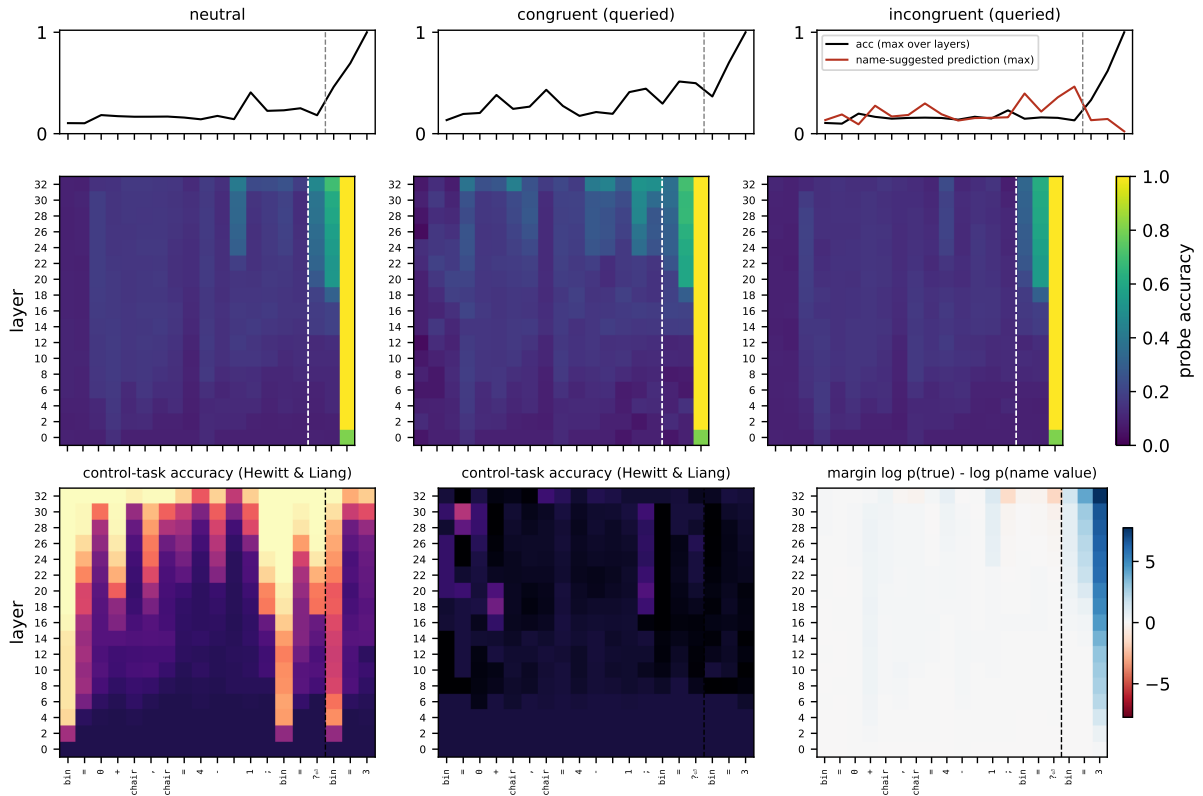


Figure 39: Llama-3.1-8B, direct regime, queried variable. Read left to right as tokens are processed, and bottom to top for later model layers. A heatmap cell gives the fraction of problems on which a separate predictor returns the correct variable digit; the colour scale runs from zero to one. Token labels illustrate one problem, while rates pool many problems. Written-calculation panels supply the correct calculation. Predictor accuracy is separate from generated-answer accuracy.

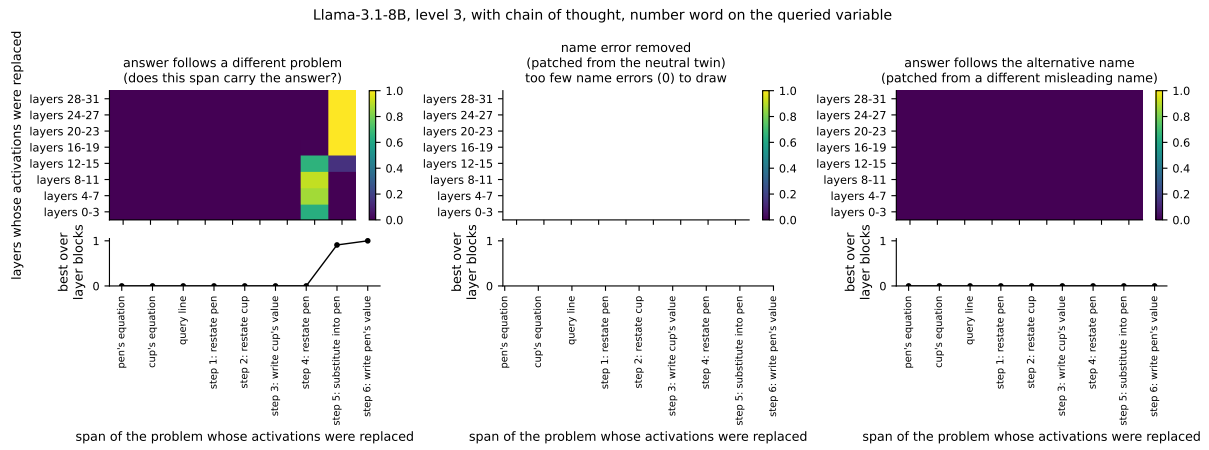


Figure 40: Llama-3.1-8B, patching grids, chain-of-thought regime, queried variable; the error-removal panel is empty for the same reason. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

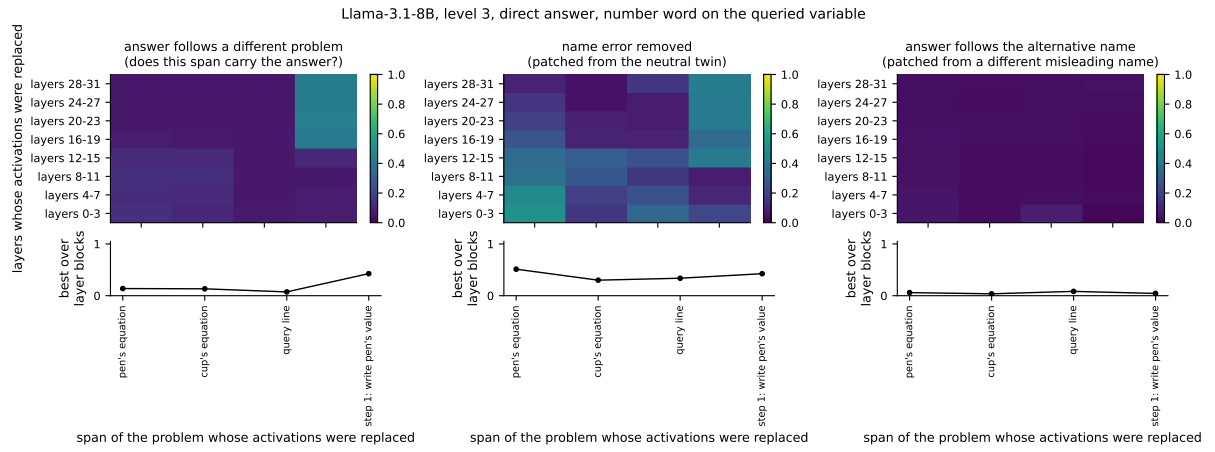


Figure 41: Llama-3.1-8B, patching grids, direct regime, queried variable. Patching replaces internal activity at the indicated position and layer with activity from a matched source problem. Removal is the fraction of baseline misleading-digit answers changed to any other digit, which can still be wrong. Damage is the fraction of baseline correct answers made wrong. These have different denominators; controls measure disruption or a different source name.

L Task competence and value readouts

Table 11 compares both variables for 9 models. Layers are selected by neutral selectivity, averaging three probe seeds. Task accuracy and paired excess name errors pool demonstration sets 7, 11 and 13; probes use set 7. The within-model difference holds overall task accuracy fixed but does not isolate a causal effect of the readout. We average the two variable readouts within each model before computing descriptive correlations. Table 12 repeats them after omitting each model. Roles are not independent observations. With this small, heterogeneous panel, correlations cannot distinguish overall competence from the role of decodable information.

M Gold-trained transfer to generated chains

We read the state before the first numeric assignment to the target in each saved generated chain. Expression operands are not counted as completed value writes; tokens that include any part of the supplied value are excluded. A fixed sample of 200 matched sets is selected by a hash independent of outcomes, shared across models, variables and demonstration sets. All additional chains with name errors and their neutral twins are evaluated separately. Neutral gold-chain probes use 10,000 separate training problems, three probe seeds and the original neutral-selectivity layer. Table 14 tests transfer to held-out neutral generated chains. A high-accuracy interpretation requires at least 90% neutral calibration in that demonstration set. Cases below that threshold yield descriptive readouts only. The augmentation is selected by observed errors and cannot estimate their population rate. Appendix N separately trains probes on generated neutral chains and tests the same held-out prefixes. Llama-3.2-3B meets the neutral transfer threshold in 6 of 6 cells, with calibration accuracy from 97.2 to 100.0%. OLMo-2-1B meets the neutral transfer threshold in 0 of 6 cells, with calibration accuracy from 19.4 to 62.2%. Name-error cases in the competent model are few; error-conditioned readouts cannot establish a general mechanism from that subset. Failed neutral calibration prevents a strong generated-chain interpretation for the exception model.

N Training probes on generated neutral chains

Gold-trained probe transfer (Appendix M) may fail because the states along generated chains differ from those along the supplied correct chain. We therefore train on freely generated neutral chains from 2,000 source problems and select layers on 400 separate validation problems. Canonical equations partition the source pool, keeping renamed versions of a computation together. The training and validation pools contain 630 and 155 computations; both exclude all test and demonstration computations. These counts distinguish independent computations from repeated name renderings. Generation is greedy with demonstration set 7 and a 128-token limit. The probe sees the hidden state before the target variable’s first numeric assignment, with every token boundary checked to exclude the value digit. Missing or unsafe boundaries are excluded and counted. We retain all usable neutral chains regardless of whether the model writes the correct value, using the executed true value as the training label. Three probe seeds use the original full-batch SGD recipe (learning rate 0.001, 10,000 epochs). States are stored in float32 and nonfinite states or fitted weights are rejected. Layers are searched at stride two, including the final layer, and chosen only by mean validation accuracy with the lowest-layer tie break. The name-identity control is trained separately and does not select layers. Table 15 reports the usable source counts and selected layers. Evaluation uses exactly the earlier fixed 200-set sample and separately reported additional name-error cases under demonstration sets 7, 11 and 13. Table 16 gives held-out neutral calibration and boundary coverage. Intervals use 4,000 canonical-computation bootstrap draws, keeping repeated names together. The 90% calibration threshold applies to the point estimate, not the lower interval endpoint. Llama passes all six calibration cells; OLMo passes none. Retraining on generated chains therefore does not validate a strong readout account of OLMo’s own errors. Table 17 retains its error readouts descriptively. Llama has only one name-error case in the primary sample and thirteen additional observed writes; these rare, selected cases cannot establish a general mechanism. Paired changes from gold-trained transfer are shown in Table 18; they compare the same eligible prefixes, not independent model samples.

Model	Role	Layer	Task	Neutral probe	Misleading probe	Excess
Gemma-3-12B-I	v1	40	100.0	100.0	100.0	0.0
Gemma-3-12B-I	v2	42	100.0	100.0	100.0	0.0
Gemma-3-4B	v1	25	100.0	100.0	100.0	0.0
Gemma-3-4B	v2	28	100.0	99.8	100.0	0.0
Gemma-3-4B-I	v1	29	100.0	100.0	99.9	0.0
Gemma-3-4B-I	v2	28	100.0	99.8	99.9	0.0
Llama-3.1-8B	v1	20	95.2	100.0	100.0	-0.1
Llama-3.1-8B	v2	22	95.2	99.4	99.2	-0.0
Llama-3.1-8B-I	v1	20	99.7	100.0	100.0	0.0
Llama-3.1-8B-I	v2	20	99.7	100.0	100.0	0.0
Llama-3.2-3B	v1	19	98.3	99.9	100.0	0.1
Llama-3.2-3B	v2	21	98.3	97.6	99.6	-0.2
Llama-3.2-3B-I	v1	18	100.0	99.9	100.0	0.0
Llama-3.2-3B-I	v2	19	100.0	99.9	99.9	0.0
OLMo-2-1B	v1	14	47.2	99.7	100.0	0.4
OLMo-2-1B	v2	0	47.2	21.8	21.8	3.2
OLMo-2-7B	v1	22	95.3	100.0	100.0	-0.1
OLMo-2-7B	v2	26	95.3	99.4	97.6	-0.0

Table 11: Level-3 arithmetic task and probe accuracies (%) and paired excess name errors (points). Neutral task accuracy is shared by the two roles. All available complete model pairs are shown; below-floor models remain descriptive.

Omitted model	Task-excess	Probe-excess	Task-probe
All models	-0.99	-1.00	+1.00
Gemma-3-12B-I	-0.99	-1.00	+1.00
Gemma-3-4B	-0.99	-1.00	+1.00
Gemma-3-4B-I	-0.99	-1.00	+1.00
Llama-3.1-8B	-0.99	-1.00	+1.00
Llama-3.1-8B-I	-0.99	-1.00	+1.00
Llama-3.2-3B	-0.99	-1.00	+1.00
Llama-3.2-3B-I	-0.99	-1.00	+1.00
OLMo-2-1B	+0.98	+0.78	+0.85
OLMo-2-7B	-0.99	-1.00	+1.00

Table 12: Descriptive Pearson correlations of model means; Spearman values and within-model variable differences are in the saved summary. No significance or causal claim is based on these coefficients.

Model	Neutral probe	Misleading probe	Excess
Gemma-3-12B-I	+0.0	+0.0	+0.0
Gemma-3-4B	-0.2	+0.0	+0.0
Gemma-3-4B-I	-0.2	+0.0	+0.0
Llama-3.1-8B	-0.6	-0.8	+0.1
Llama-3.1-8B-I	+0.0	+0.0	+0.0
Llama-3.2-3B	-2.2	-0.4	-0.3
Llama-3.2-3B-I	+0.1	-0.1	+0.0
OLMo-2-1B	-77.9	-78.2	+2.7
OLMo-2-7B	-0.5	-2.3	+0.1

Table 13: Intermediate minus queried variable, in percentage points. Overall neutral task accuracy is identical within each model. These differences are descriptive comparisons between variable positions.

Model	Set	Role	Neutral coverage	Calibration	Sample error probe	Extra errors n	Extra error probe
OLMo-2-1B	7	v1	194/200	62.2 [55.2, 68.7]	0.0 [0.0, 0.0]	69	1.4 [0.0, 3.9]
OLMo-2-1B	7	v2	186/200	19.4 [14.0, 25.3]	54.5 [27.3, 81.8]	82	65.9 [56.1, 75.6]
OLMo-2-1B	11	v1	187/200	55.3 [48.3, 62.6]	0.0 [0.0, 0.0]	59	5.6 [0.6, 12.4]
OLMo-2-1B	11	v2	129/200	21.7 [14.7, 28.7]	100.0 [100.0, 100.0]	21	66.7 [47.6, 85.7]
OLMo-2-1B	13	v1	193/200	52.7 [45.6, 59.8]	27.5 [7.8, 49.0]	159	10.7 [6.5, 15.3]
OLMo-2-1B	13	v2	124/200	24.2 [16.9, 32.3]	100.0 [100.0, 100.0]	30	60.0 [43.3, 76.7]
Llama-3.2-3B	7	v1	200/200	98.0 [96.0, 99.5]	0.0 [0.0, 0.0]	8	12.5 [0.0, 37.5]
Llama-3.2-3B	7	v2	200/200	97.2 [94.8, 99.0]	–	0	–
Llama-3.2-3B	11	v1	199/200	100.0 [100.0, 100.0]	–	0	–
Llama-3.2-3B	11	v2	200/200	100.0 [100.0, 100.0]	–	0	–
Llama-3.2-3B	13	v1	199/200	99.5 [98.5, 100.0]	–	5	0.0 [0.0, 0.0]
Llama-3.2-3B	13	v2	200/200	99.5 [98.5, 100.0]	–	0	–

Table 14: Generated-chain true-digit accuracy (%) [95% interval]. Sample error probe conditions on name errors in the fixed sample; extra name-error cases exclude that sample. Dashes indicate no eligible cases. Saved summaries record all missing or excluded boundaries and calibration decisions.

Model	Role	Train n	Validation n	Layer	Validation (%)	Name ID (%)
OLMo-2-1B-I	v1	1924	390	14	47.9	41.6
OLMo-2-1B-I	v2	1877	376	16	87.8	86.6
Llama-3.2-3B	v1	1994	398	22	97.5	26.8
Llama-3.2-3B	v2	2000	400	20	99.2	28.6

Table 15: Probe source rows with usable generated boundaries, selected layers and validation scores. Source cohorts are computation-disjoint; usable counts vary by model and variable. Validation scores select layers and are not test estimates.

Model	Role	Set	Neutral coverage	Calibration (%)	Name ID (%)	Pass
OLMo-2-1B-I	v1	7	194/200	61.3 [54.1, 68.6]	37.3 [31.2, 43.4]	no
OLMo-2-1B-I	v1	11	187/200	52.8 [45.1, 60.3]	32.4 [26.6, 38.5]	no
OLMo-2-1B-I	v1	13	193/200	50.3 [43.0, 57.4]	35.2 [29.1, 41.6]	no
OLMo-2-1B-I	v2	7	186/200	78.0 [71.5, 83.9]	81.9 [76.9, 86.7]	no
OLMo-2-1B-I	v2	11	129/200	80.1 [72.8, 86.8]	56.6 [49.6, 63.3]	no
OLMo-2-1B-I	v2	13	124/200	71.8 [64.2, 78.4]	46.8 [37.8, 55.6]	no
Llama-3.2-3B	v1	7	200/200	98.5 [96.7, 100.0]	18.5 [13.8, 23.7]	yes
Llama-3.2-3B	v1	11	199/200	100.0 [100.0, 100.0]	13.9 [9.5, 18.5]	yes
Llama-3.2-3B	v1	13	199/200	97.2 [94.2, 99.5]	16.9 [12.3, 21.8]	yes
Llama-3.2-3B	v2	7	200/200	96.3 [93.6, 98.6]	28.8 [22.7, 35.0]	yes
Llama-3.2-3B	v2	11	200/200	100.0 [100.0, 100.0]	17.2 [12.4, 22.1]	yes
Llama-3.2-3B	v2	13	200/200	97.7 [95.5, 99.3]	17.8 [13.3, 22.8]	yes

Table 16: Held-out primary neutral generated chains. Coverage counts valid pre-value boundaries over all primary neutral rows. Accuracy intervals are 95% canonical-computation bootstrap intervals; pass means the calibration point estimate meets 90%.

Model	Role	Set	Sample n	Sample accuracy	Extra n	Extra accuracy
OLMo-2-1B-I	v1	7	8	4.2 [0.0, 12.5]	69	2.9 [0.0, 7.0]
OLMo-2-1B-I	v1	11	7	19.0 [0.0, 38.1]	59	10.7 [4.2, 18.8]
OLMo-2-1B-I	v1	13	17	21.6 [7.8, 39.2]	159	13.6 [8.6, 19.4]
OLMo-2-1B-I	v2	7	11	33.3 [15.2, 51.5]	82	18.7 [11.4, 27.2]
OLMo-2-1B-I	v2	11	2	0.0 [0.0, 0.0]	21	17.5 [3.3, 31.8]
OLMo-2-1B-I	v2	13	1	0.0 [0.0, 0.0]	30	25.6 [13.7, 39.1]
Llama-3.2-3B	v1	7	1	0.0 [0.0, 0.0]	8	12.5 [0.0, 42.9]
Llama-3.2-3B	v1	11	0	–	0	–
Llama-3.2-3B	v1	13	0	–	5	0.0 [0.0, 0.0]
Llama-3.2-3B	v2	7	0	–	0	–
Llama-3.2-3B	v2	11	0	–	0	–
Llama-3.2-3B	v2	13	0	–	0	–

Table 17: True-digit accuracy (%) [95% interval] on actual name errors, with the fixed sample and additional error-selected cases separated. Extra cases exclude the primary sample. Dashes denote empty strata. OLMo fails neutral calibration, so its readouts are descriptive.

Model	Role	Set	Neutral change	Sample error change	Extra error change
OLMo-2-1B-I	v1	7	-0.9 [-4.9, 3.4]	4.2 [0.0, 12.5]	1.4 [-1.0, 5.3]
OLMo-2-1B-I	v1	11	-2.5 [-7.7, 2.7]	19.0 [0.0, 38.1]	5.1 [-0.6, 12.1]
OLMo-2-1B-I	v1	13	-2.4 [-7.2, 2.2]	-5.9 [-19.6, 5.9]	2.9 [-0.8, 6.7]
OLMo-2-1B-I	v2	7	58.6 [49.5, 67.5]	-21.2 [-57.6, 12.2]	-47.2 [-63.3, -30.1]
OLMo-2-1B-I	v2	11	58.4 [47.9, 67.9]	-100.0 [-100.0, -100.0]	-49.2 [-71.4, -20.8]
OLMo-2-1B-I	v2	13	47.6 [37.7, 57.1]	-100.0 [-100.0, -100.0]	-34.4 [-55.6, -13.3]
Llama-3.2-3B	v1	7	0.5 [0.0, 1.6]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Llama-3.2-3B	v1	11	0.0 [0.0, 0.0]	—	—
Llama-3.2-3B	v1	13	-2.3 [-5.1, -0.3]	—	0.0 [0.0, 0.0]
Llama-3.2-3B	v2	7	-0.8 [-2.9, 0.9]	—	—
Llama-3.2-3B	v2	11	0.0 [0.0, 0.0]	—	—
Llama-3.2-3B	v2	13	-1.8 [-3.7, -0.5]	—	—

Table 18: Paired accuracy changes in percentage points [95% interval] from gold-trained transfer to generated-neutral training on the same valid prefixes. Error-selection and failed neutral calibration limit the interpretation.